

# Update – Hardware

JC Vaught

Aug 19, 2025

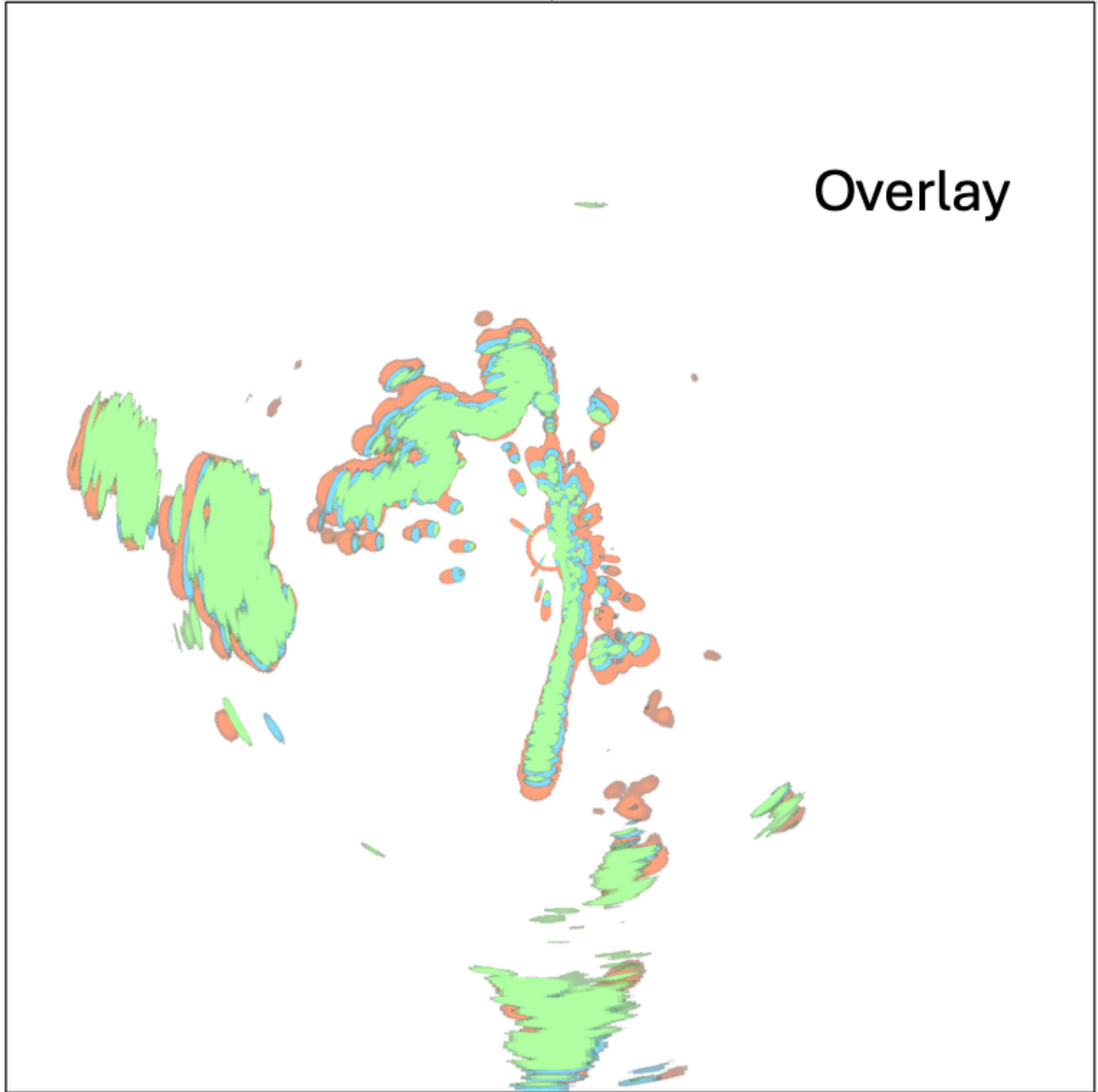
# Exec Summary

- Radar
  - Collected more data at lake
  - Created code to collect multi-pulse-length HDR combination
    - Results next week
- AI/ML
  - Tested FFT + Hough TF analysis for object tracking
  - Tried Background subtraction
  - Tested zero shot image segmentation(fail)
  - Built custom MAE implementation 5mil - 2.3b params
  - Tested masking ratios, model size on performance

# Radar Overview



Overlay



# Object Tracking - Classic

## 3D FFT SIGNAL PROCESSING

- Temporal sequence 3D data cube creation
- Fast Fourier Transform frequency analysis
- Advanced clutter filtering and spectrum visualization

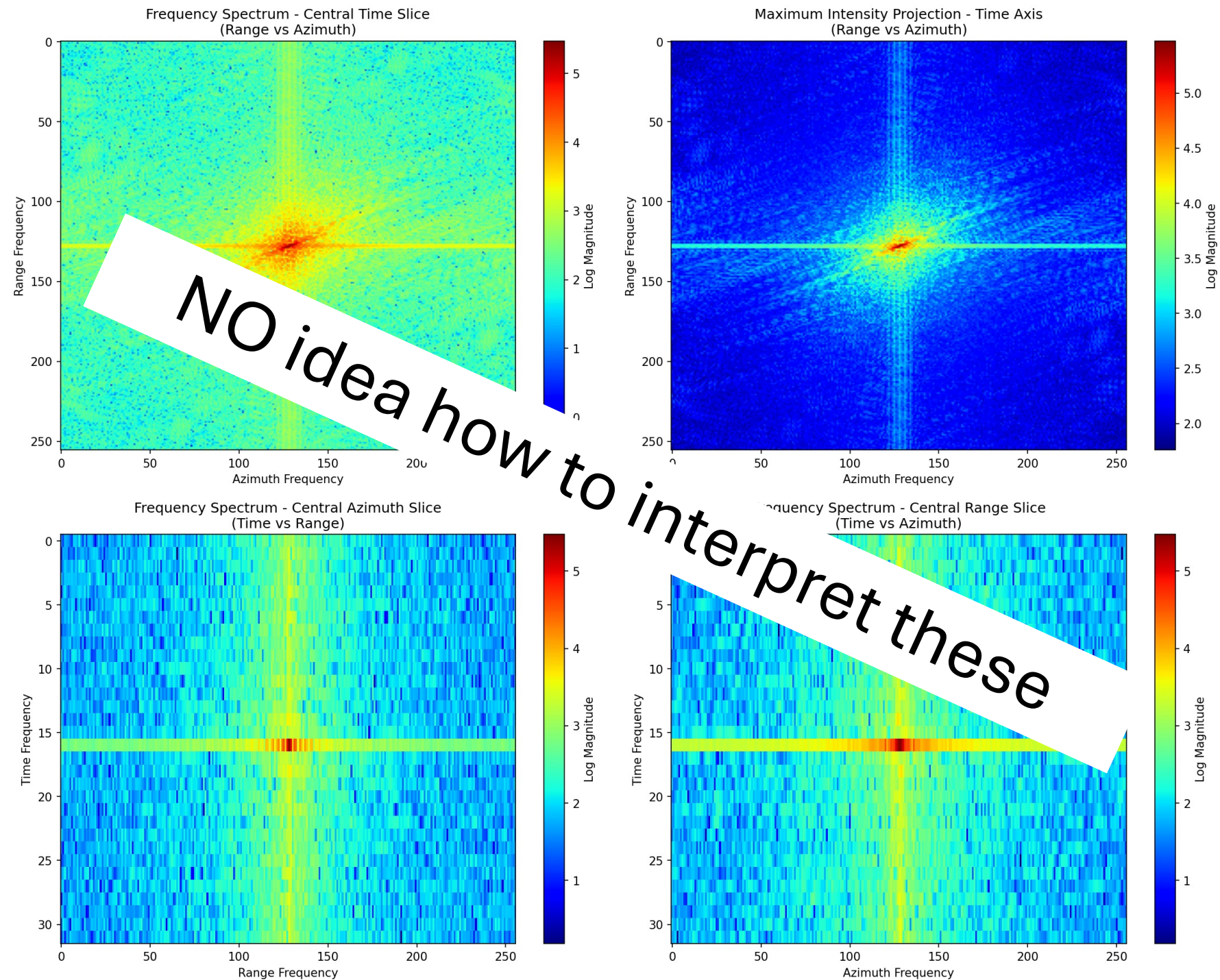
## DETECTION ALGORITHMS

- Background subtraction with adaptive thresholding
- Frame-to-frame motion detection analysis
- Intensity-based radar reflector identification
- Hough transform for linear track detection

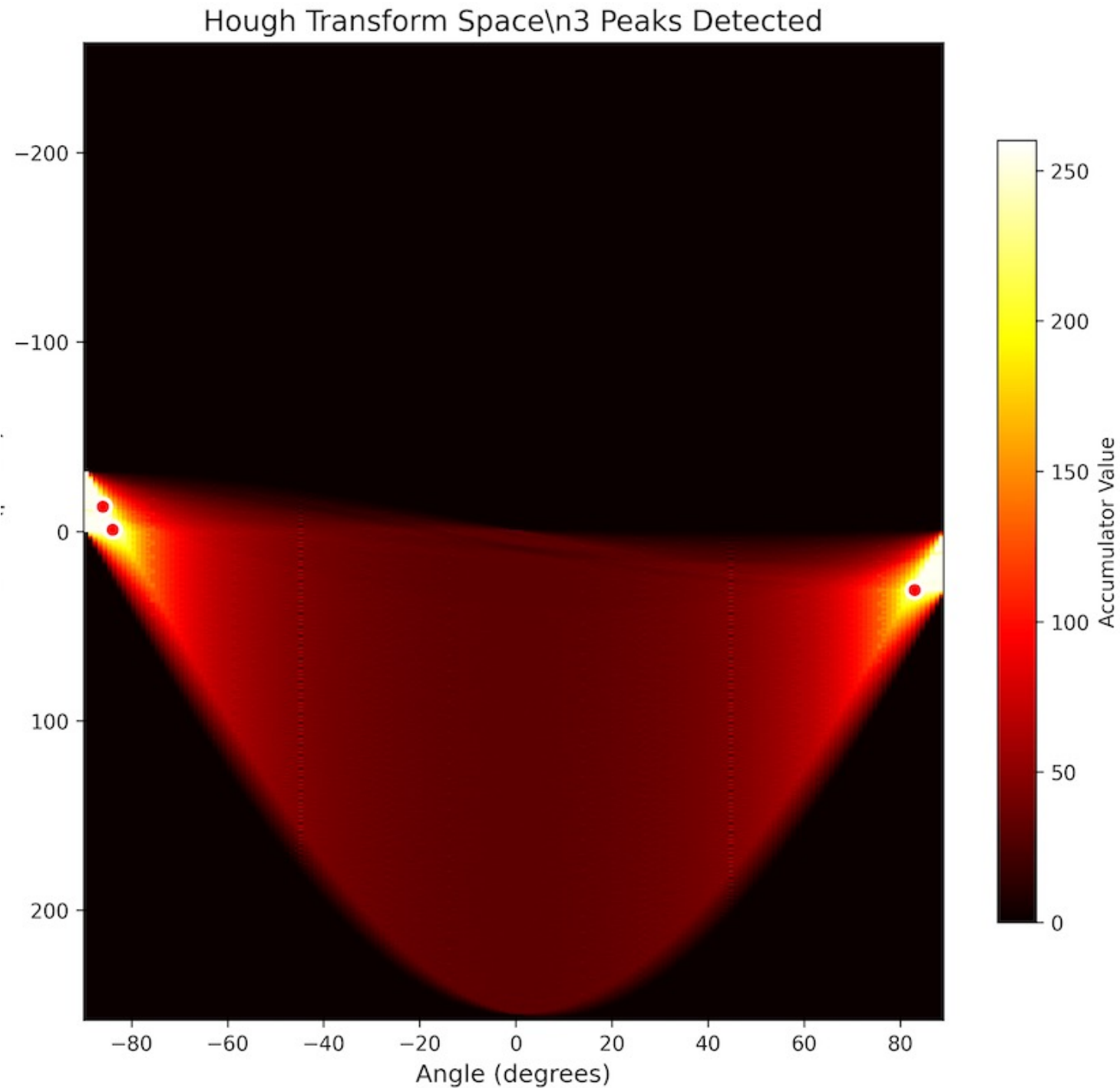
## TRACKING & VISUALIZATION

- Multi-object tracking with bounding boxes
- Track persistence filtering and association
- Motion prediction and trajectory analysis

# Object Tracking - FFT



# Object Tracking – Hough Transformation

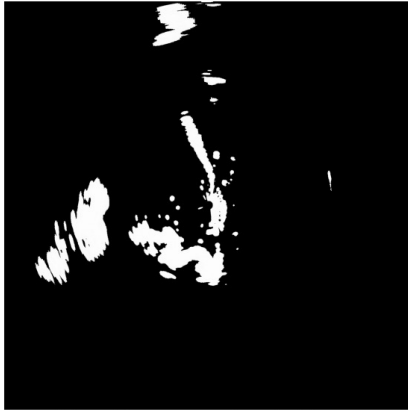


# Object Tracking – Intensity Based



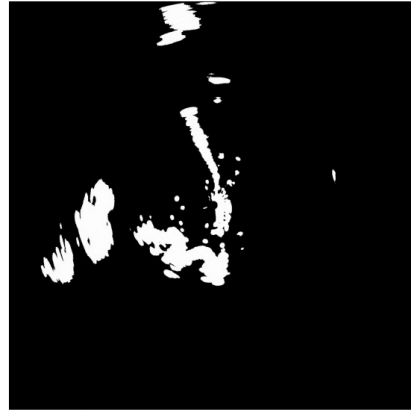
# Intensity

Frame 100



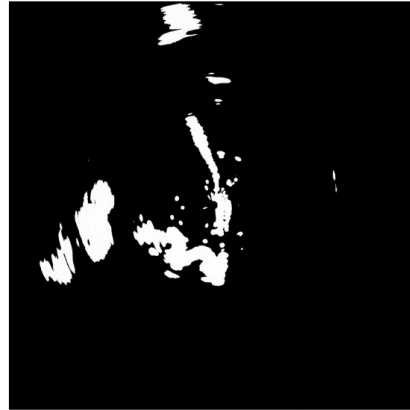
Bright Objects: 29\nThreshold: 0.992

Frame 200



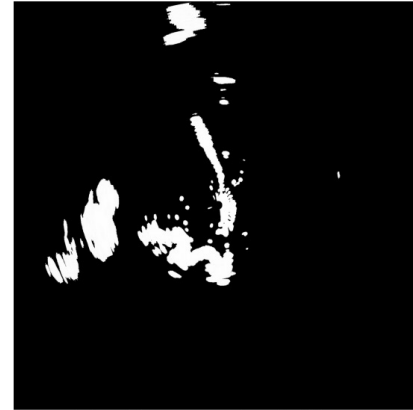
Bright Objects: 32\nThreshold: 0.992

Frame 300



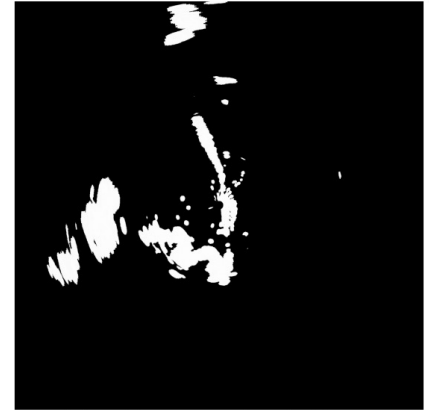
Bright Objects: 0\nThreshold: 0.996

Frame 400

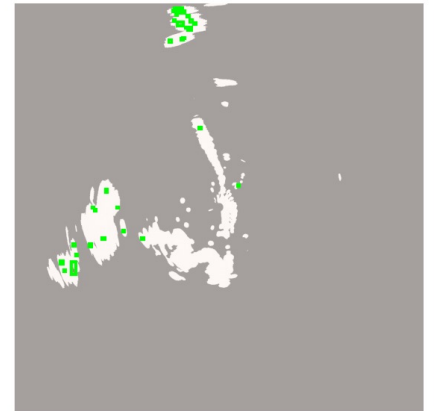
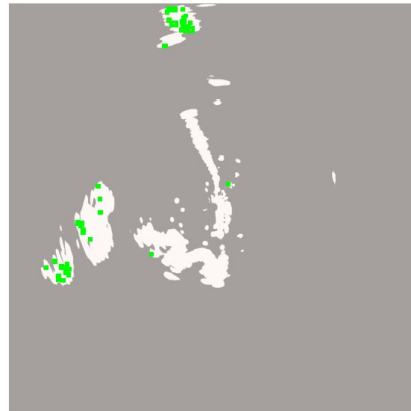
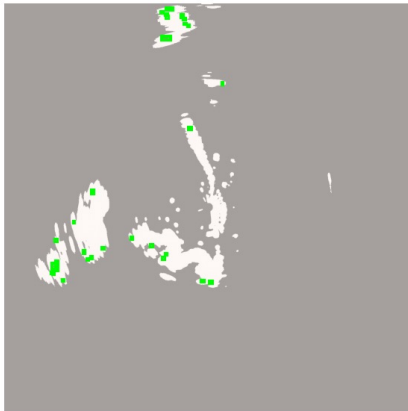


Bright Objects: 0\nThreshold: 0.996

Frame 500



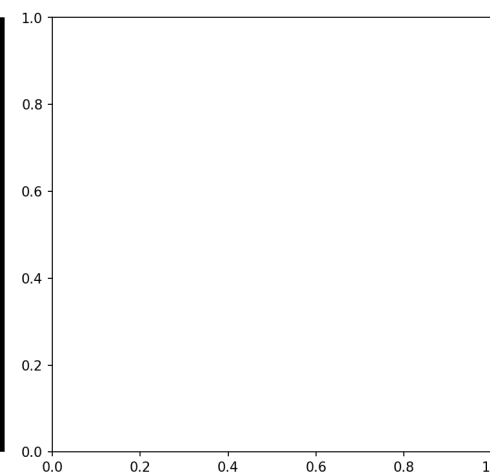
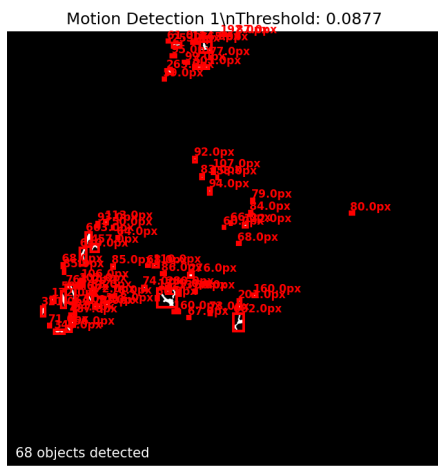
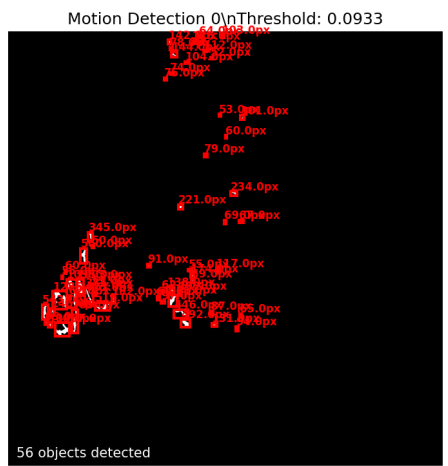
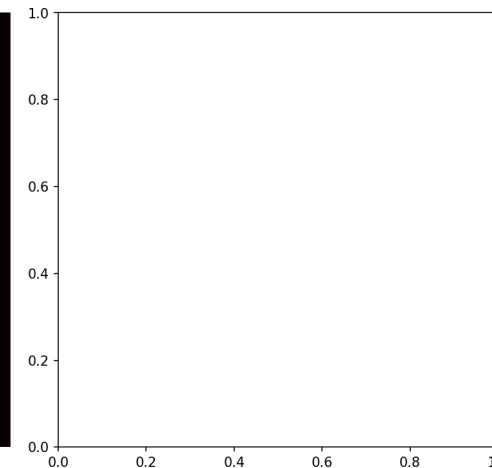
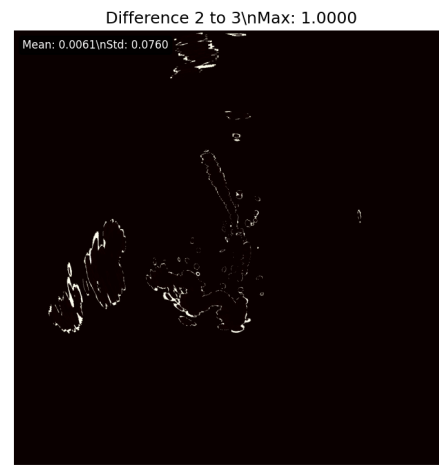
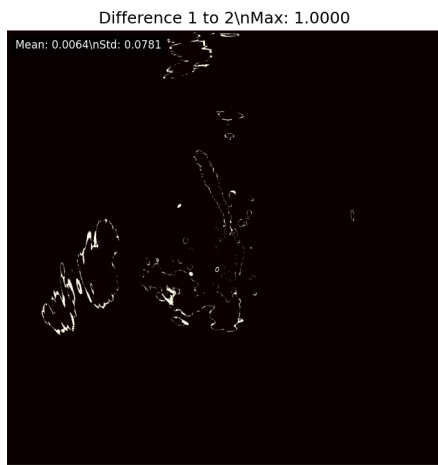
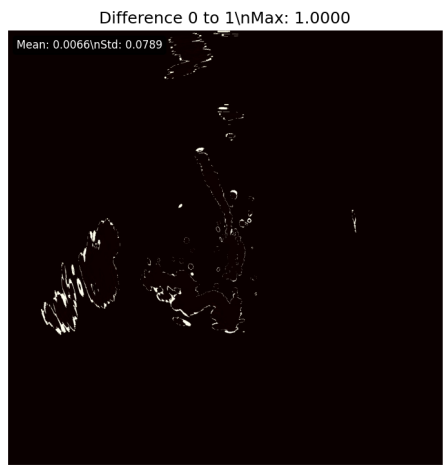
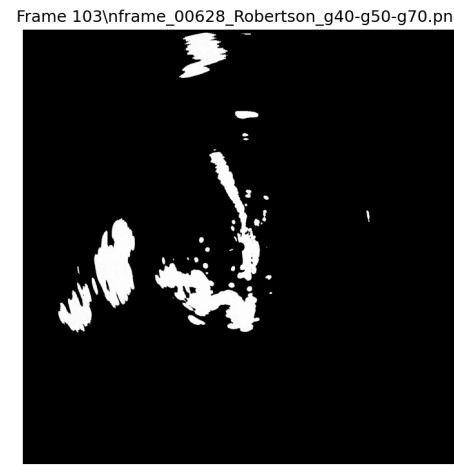
Bright Objects: 29\nThreshold: 0.992



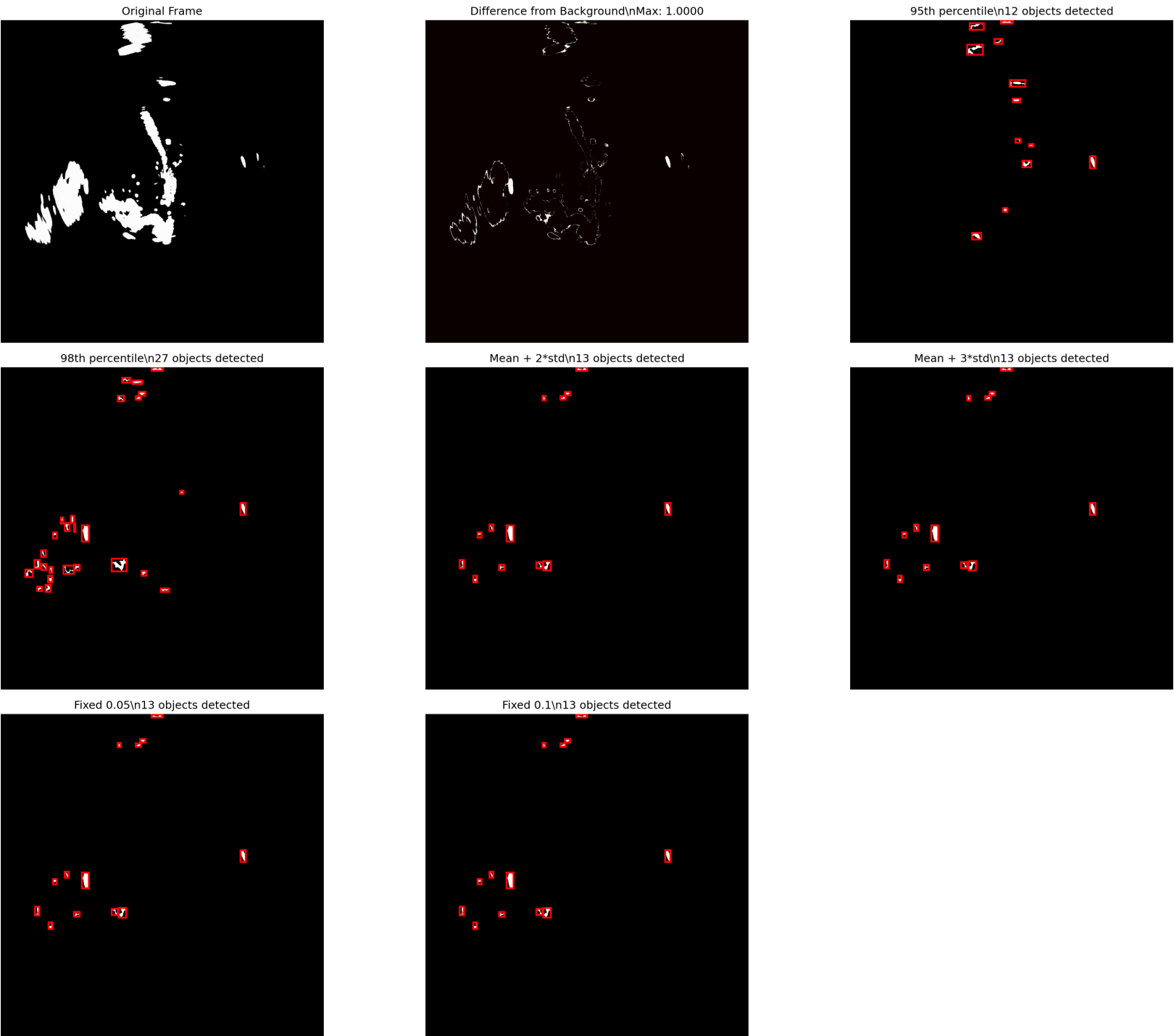
# Object Tracking – Motion Based



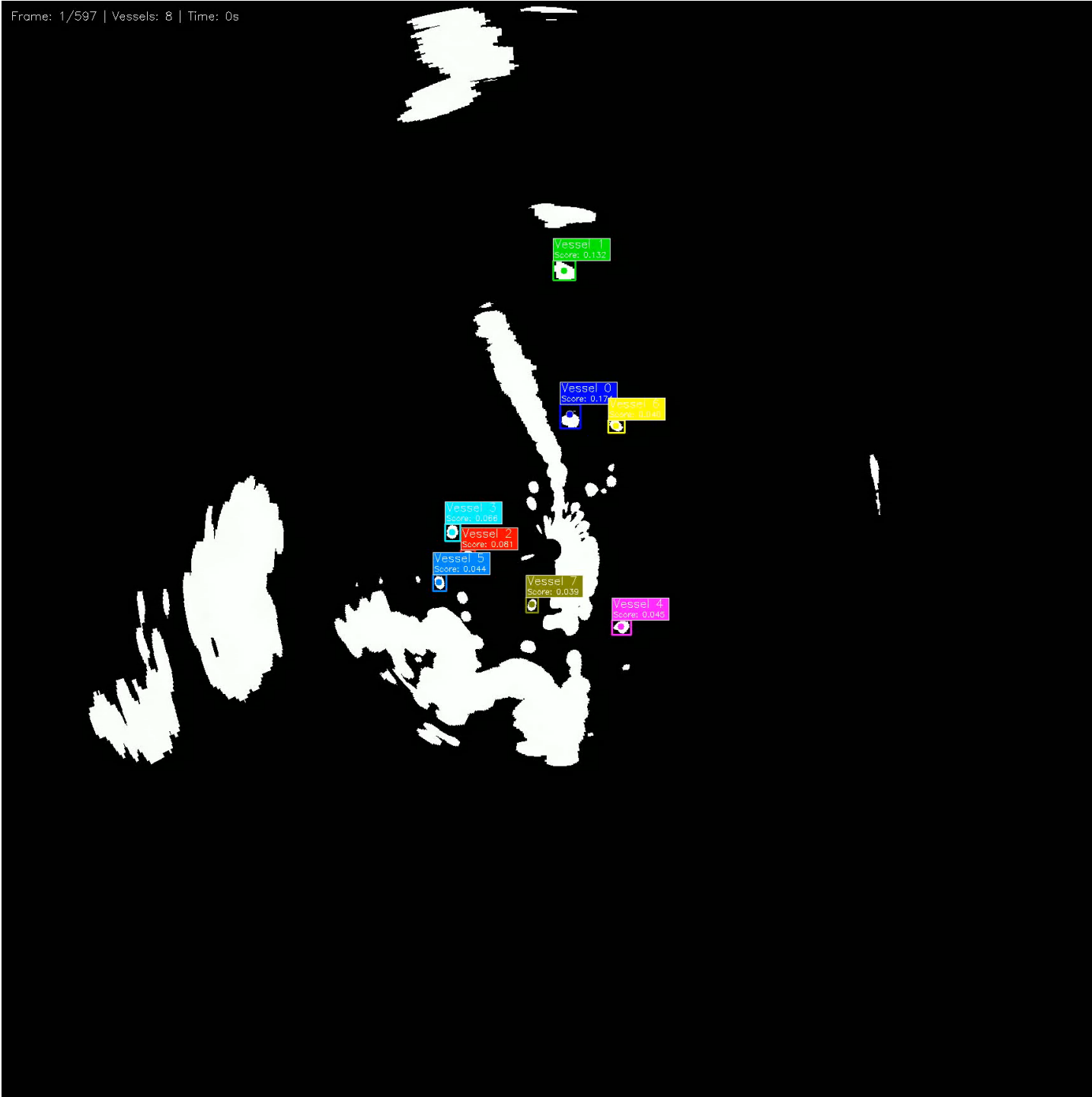
# Motion



# Motion



# Object Tracking – Don't Remember

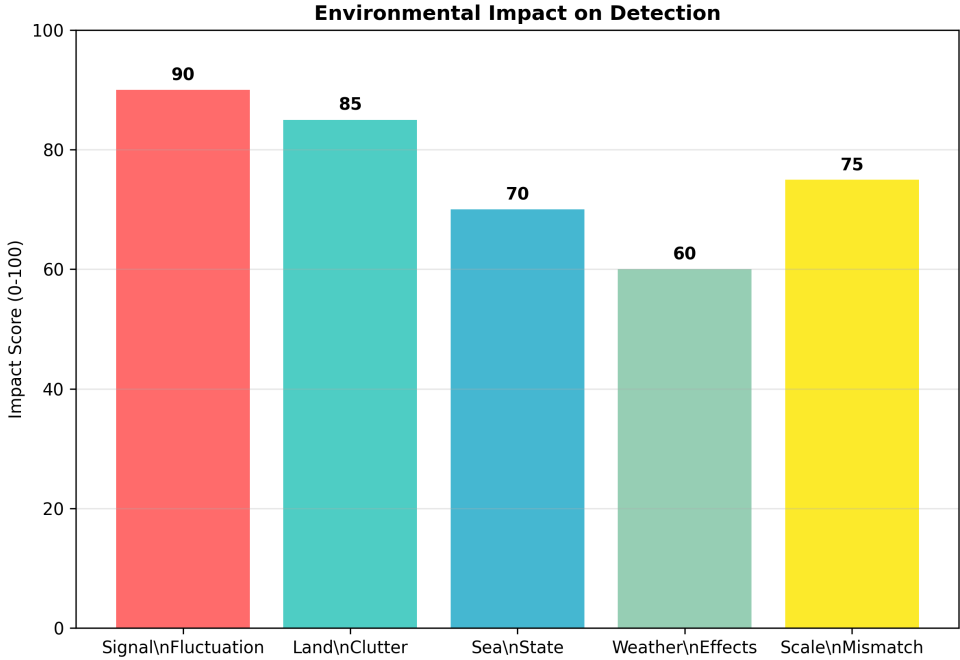
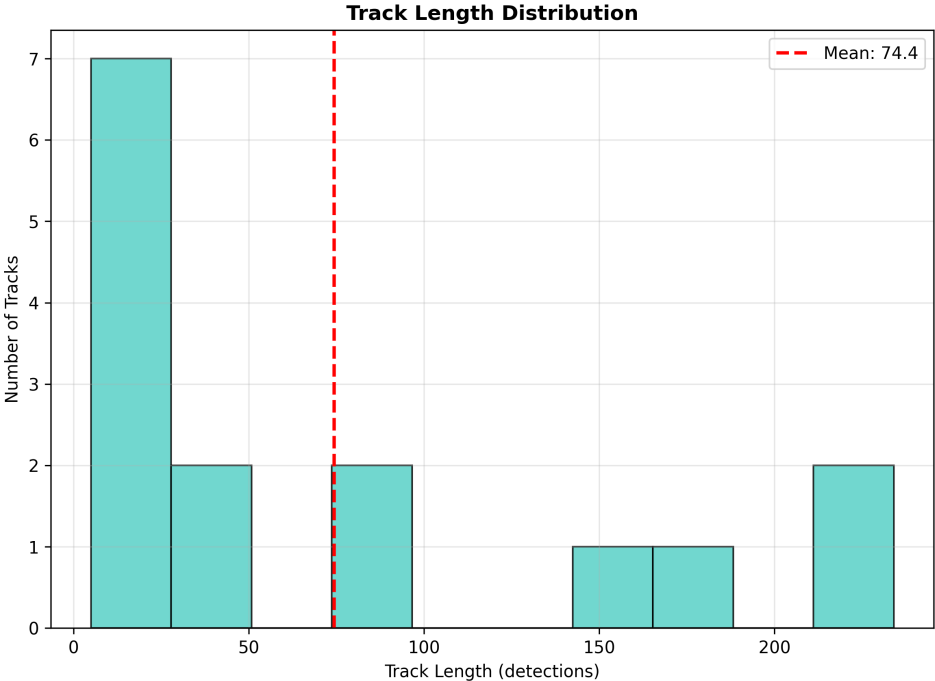
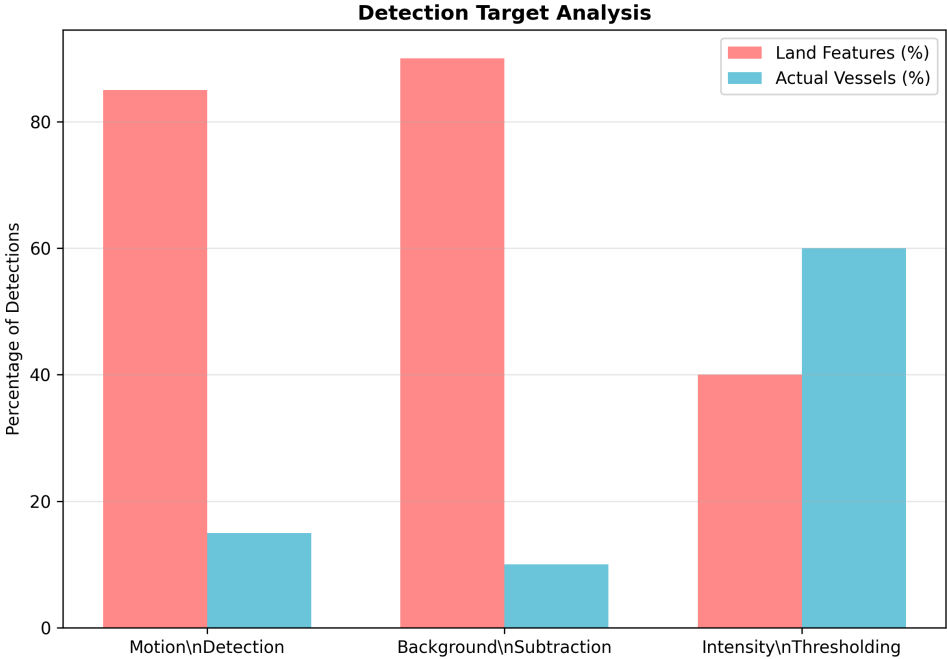
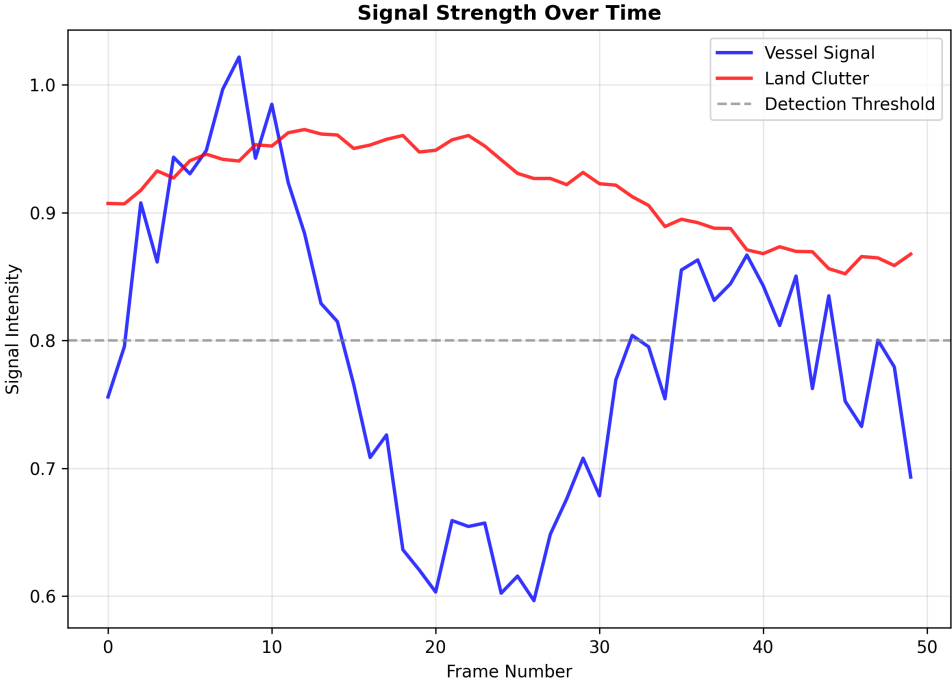


# Object Tracking – Don't Remember

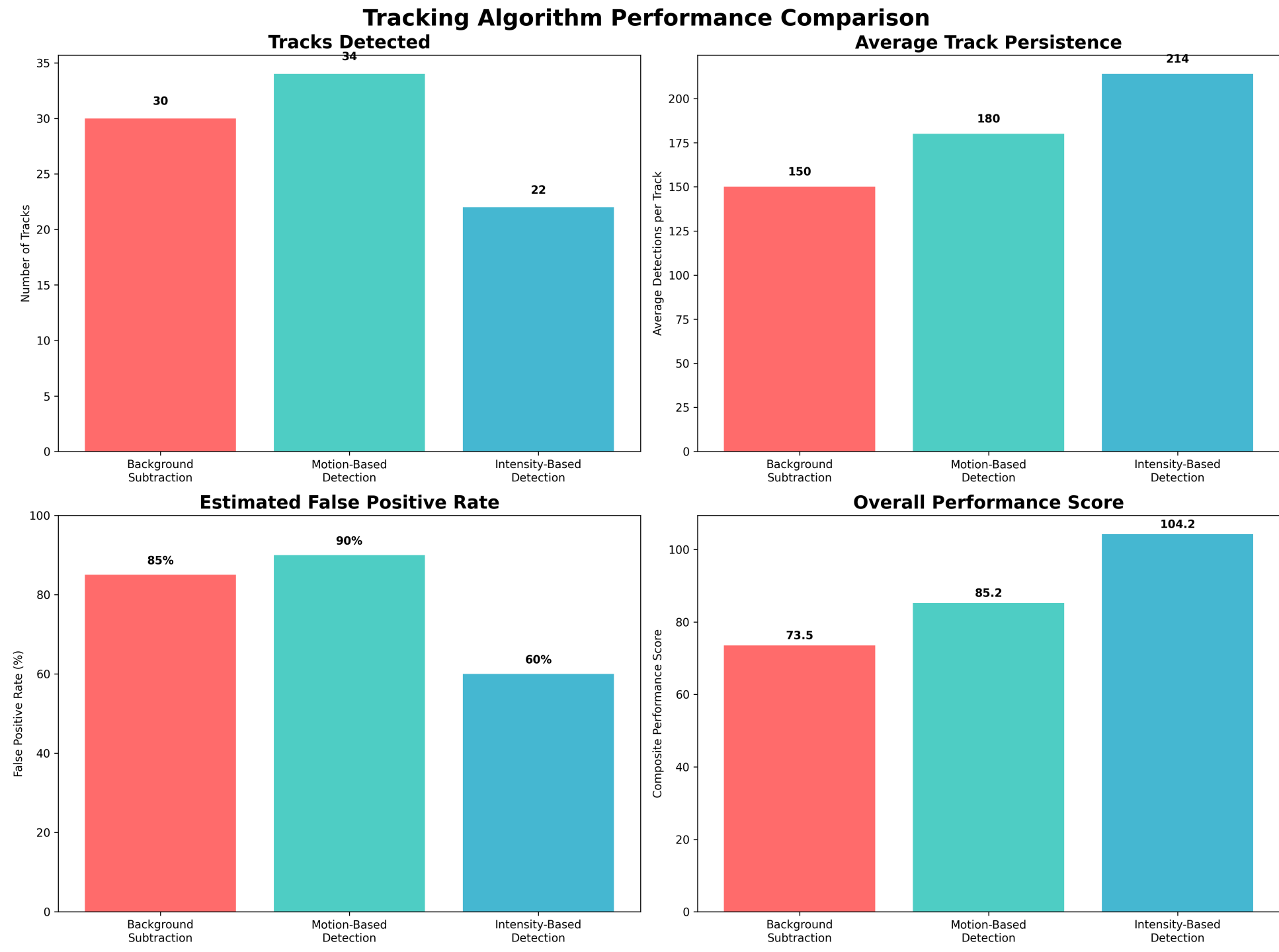


# summary

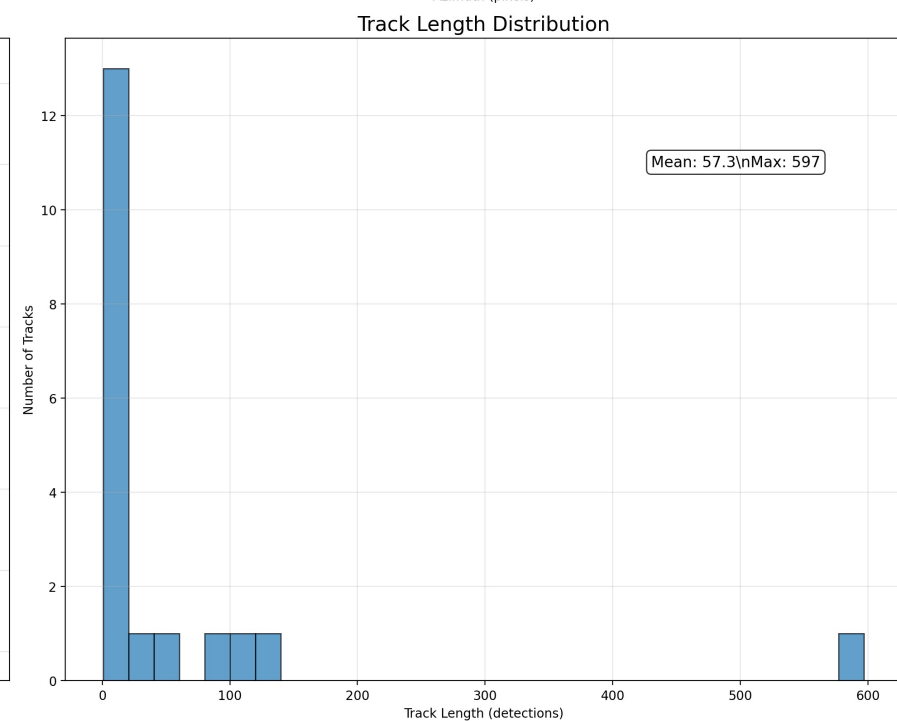
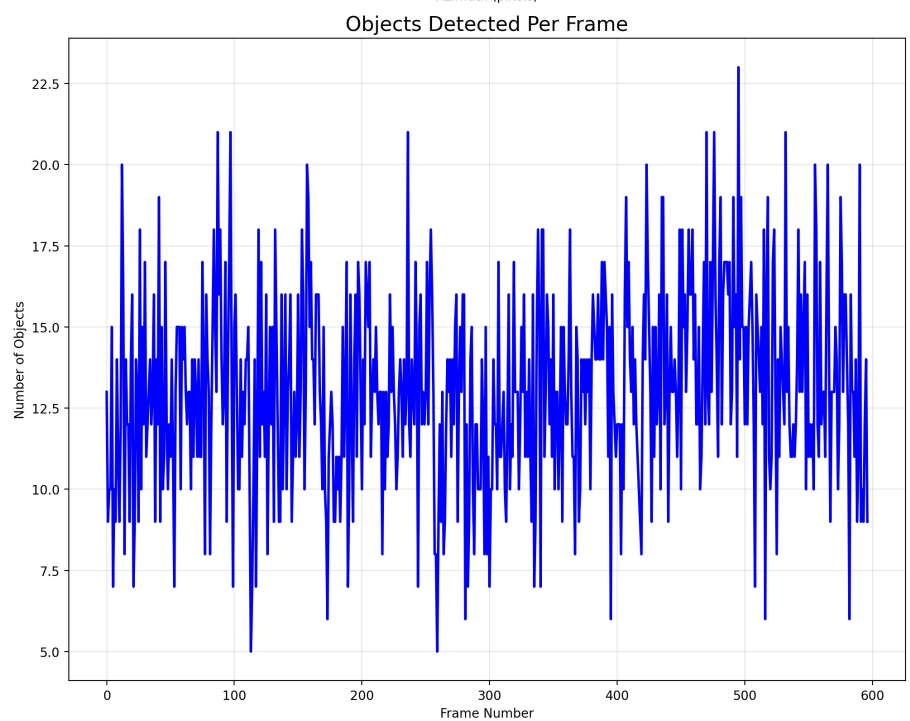
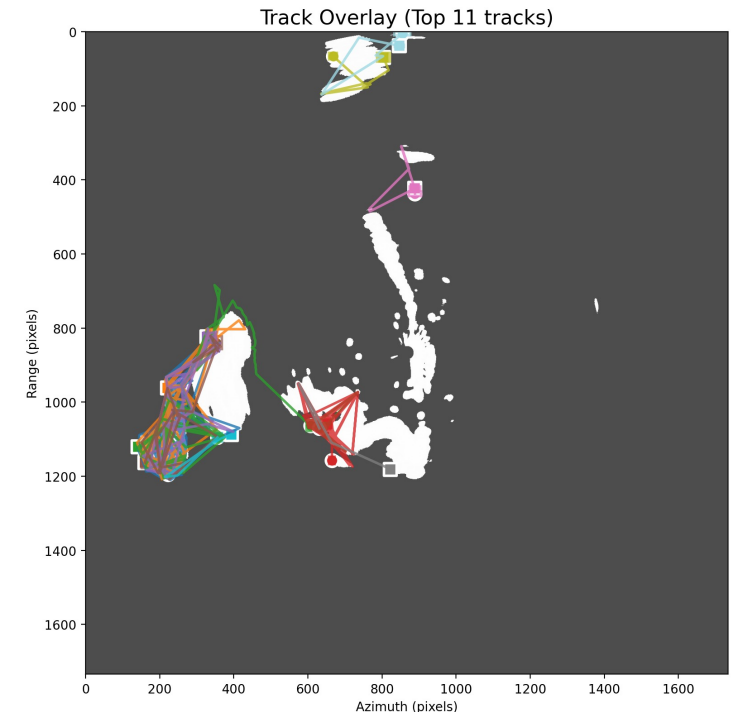
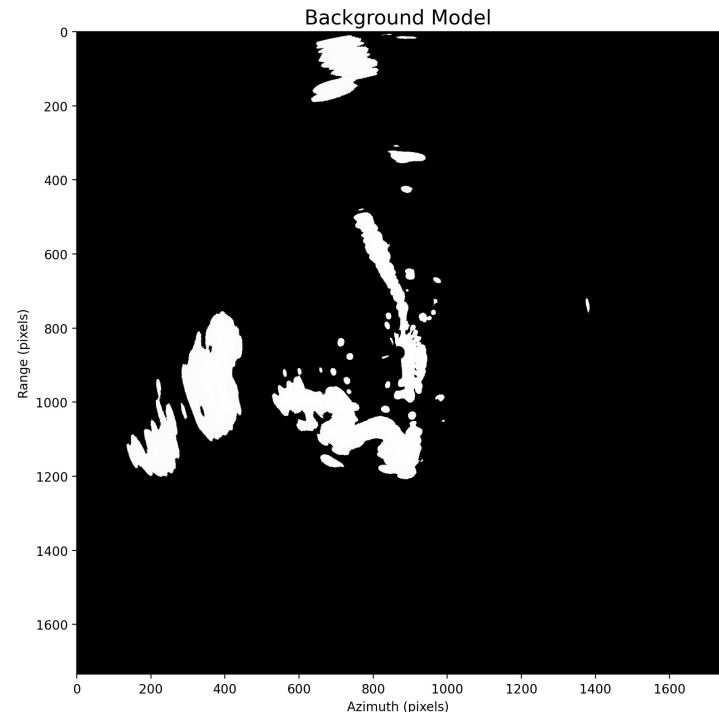
## Radar Detection Challenges Analysis



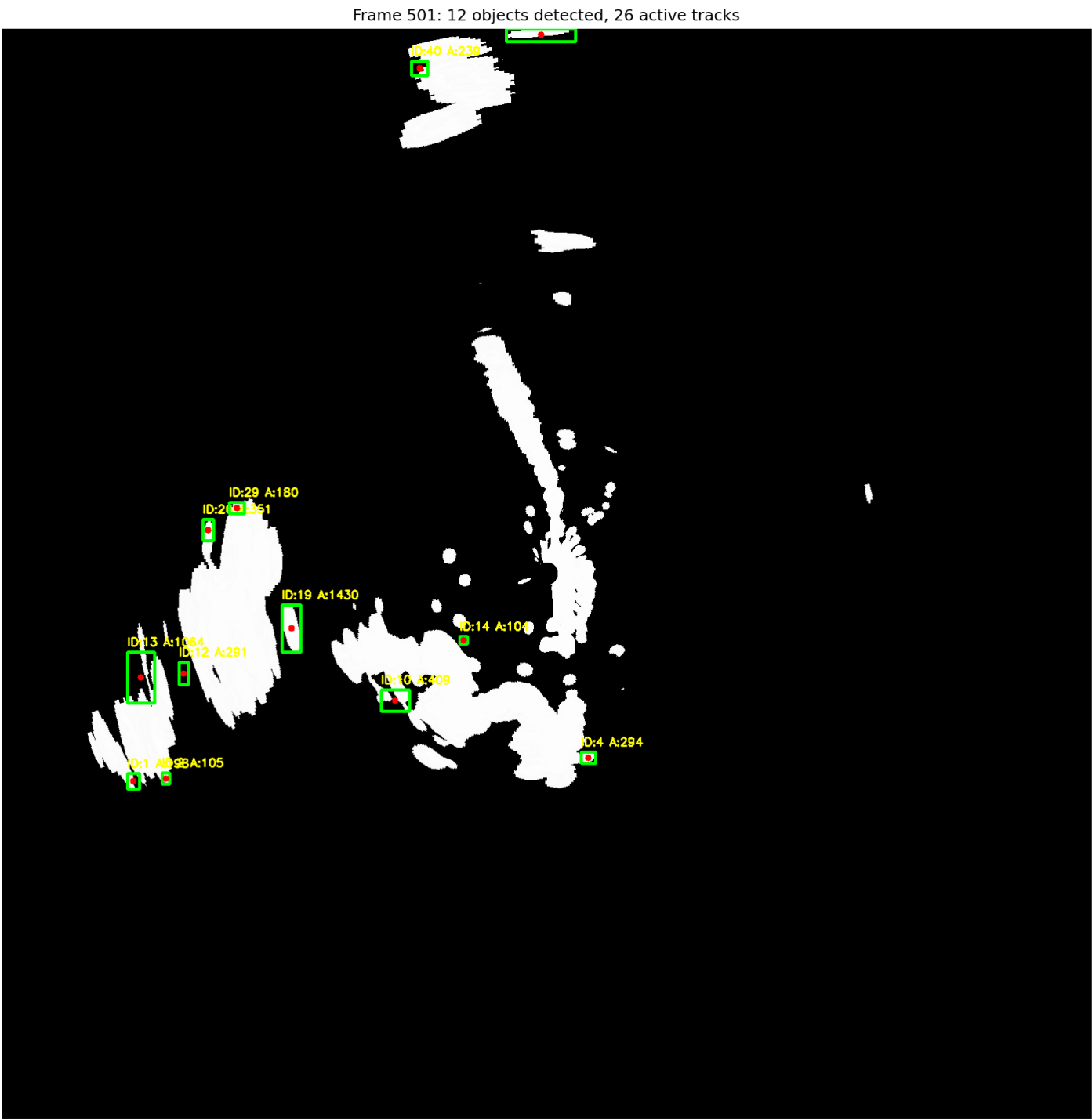
# summary



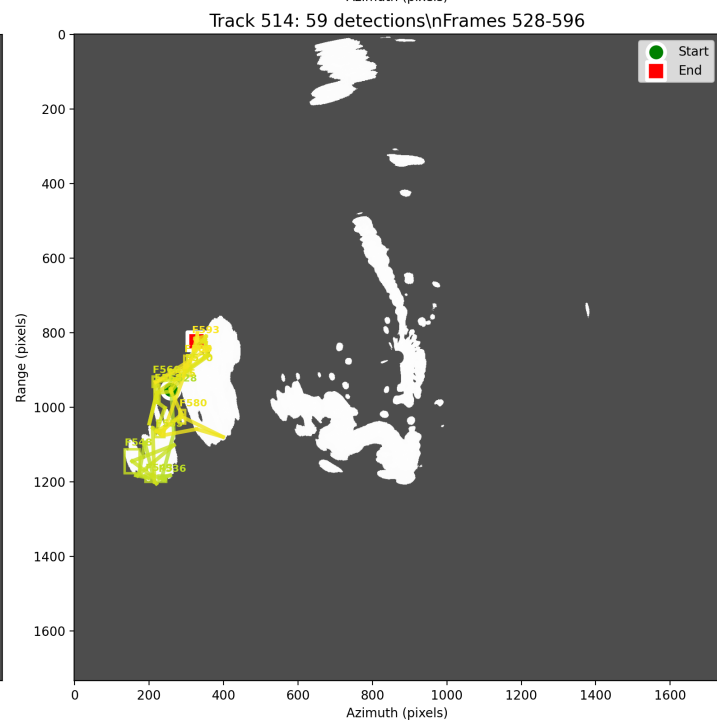
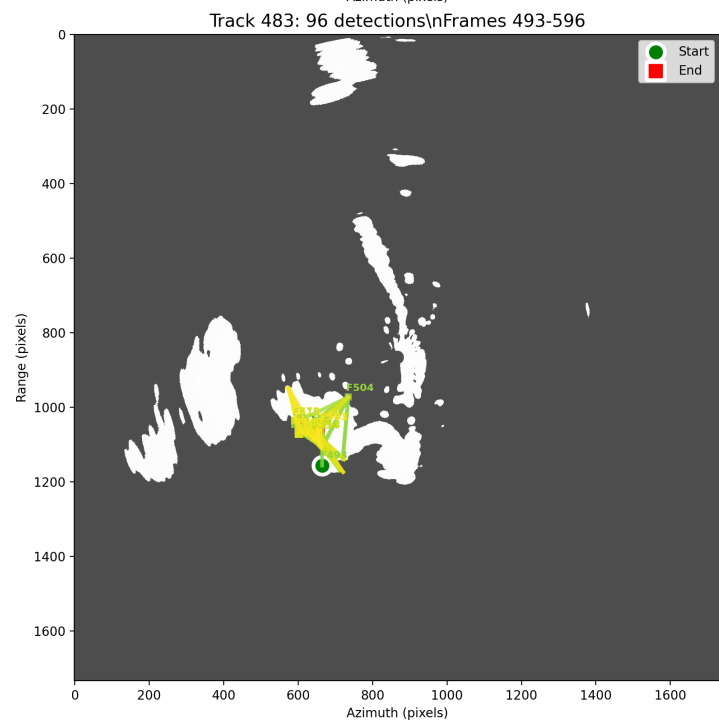
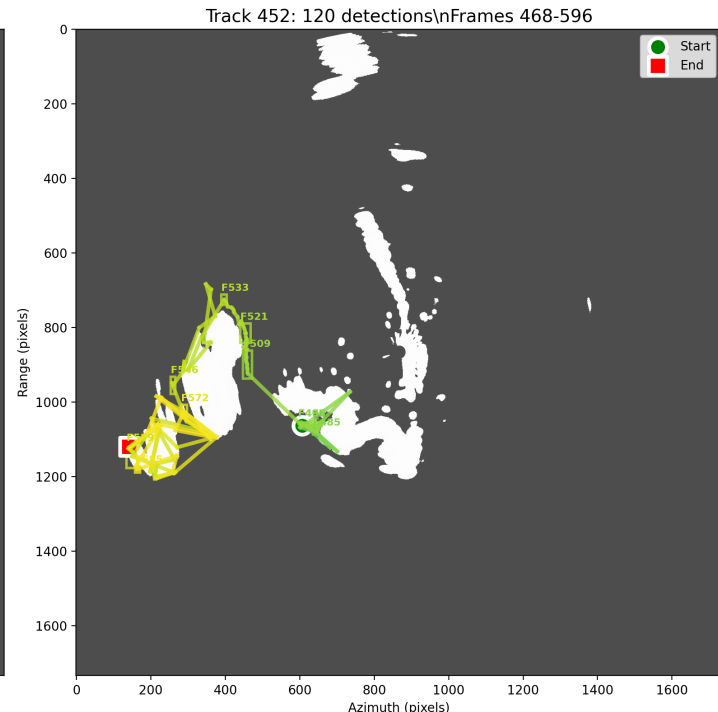
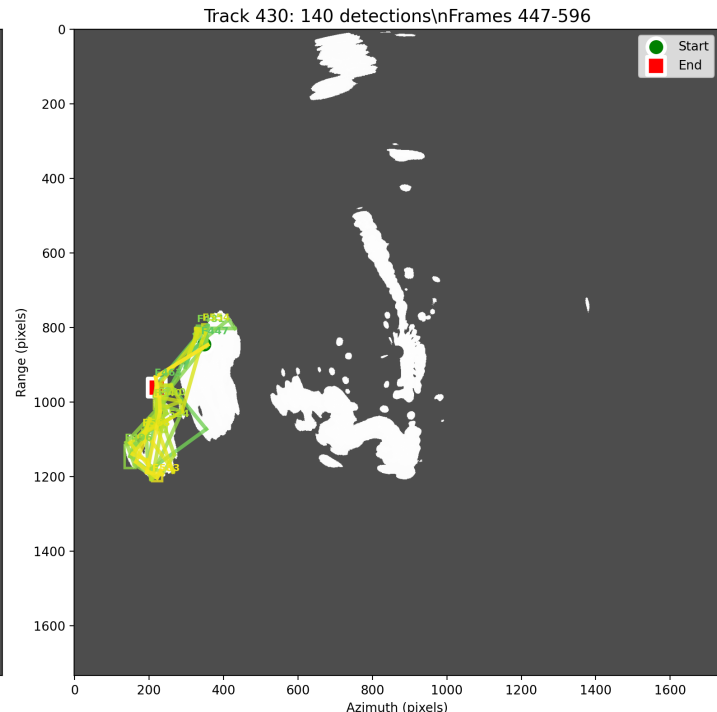
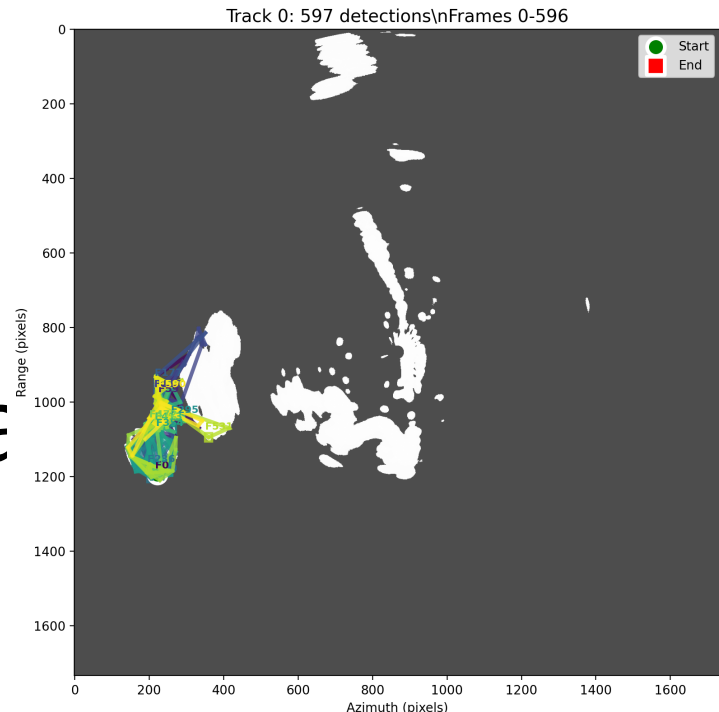
# summary



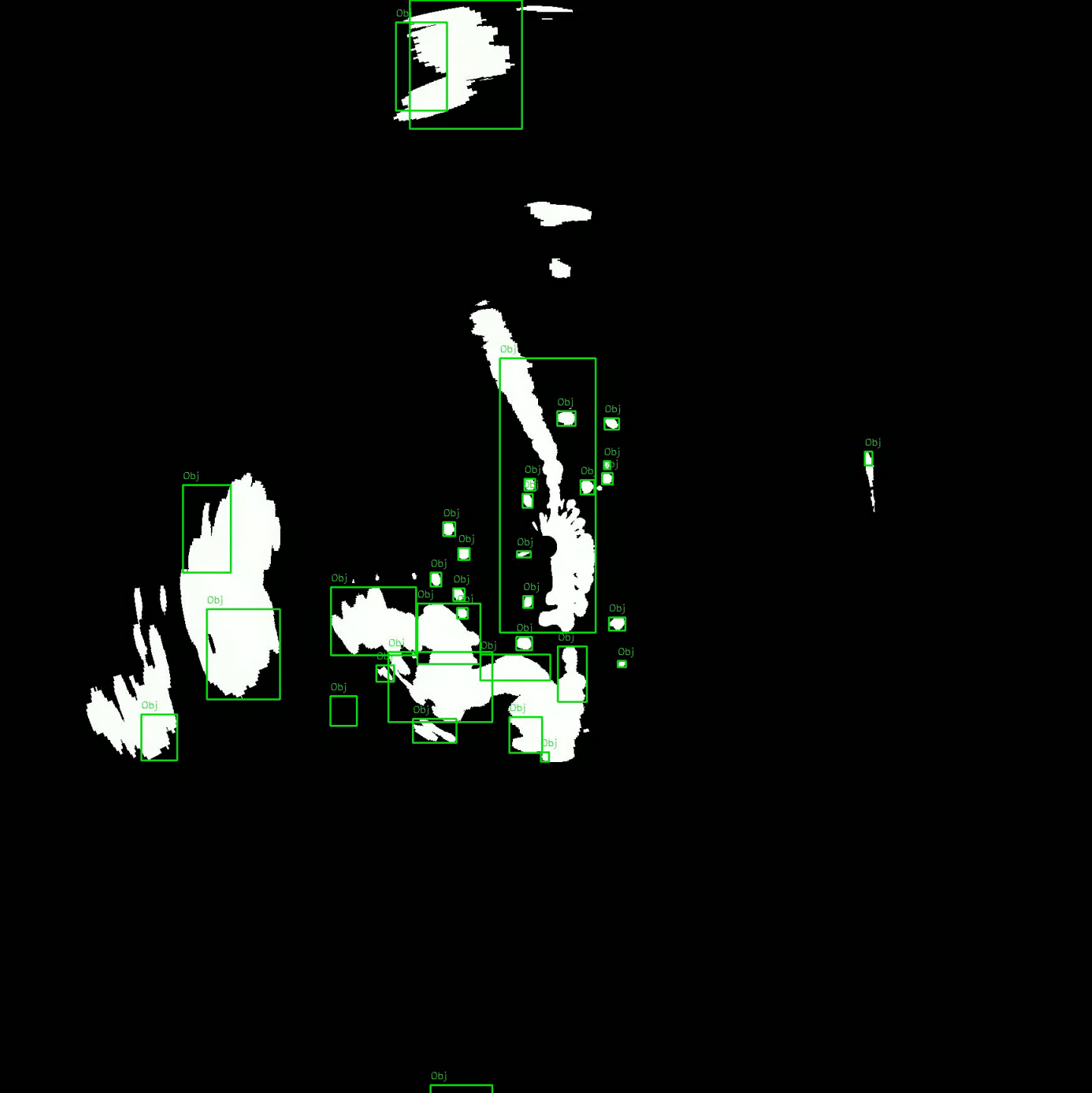
# summary



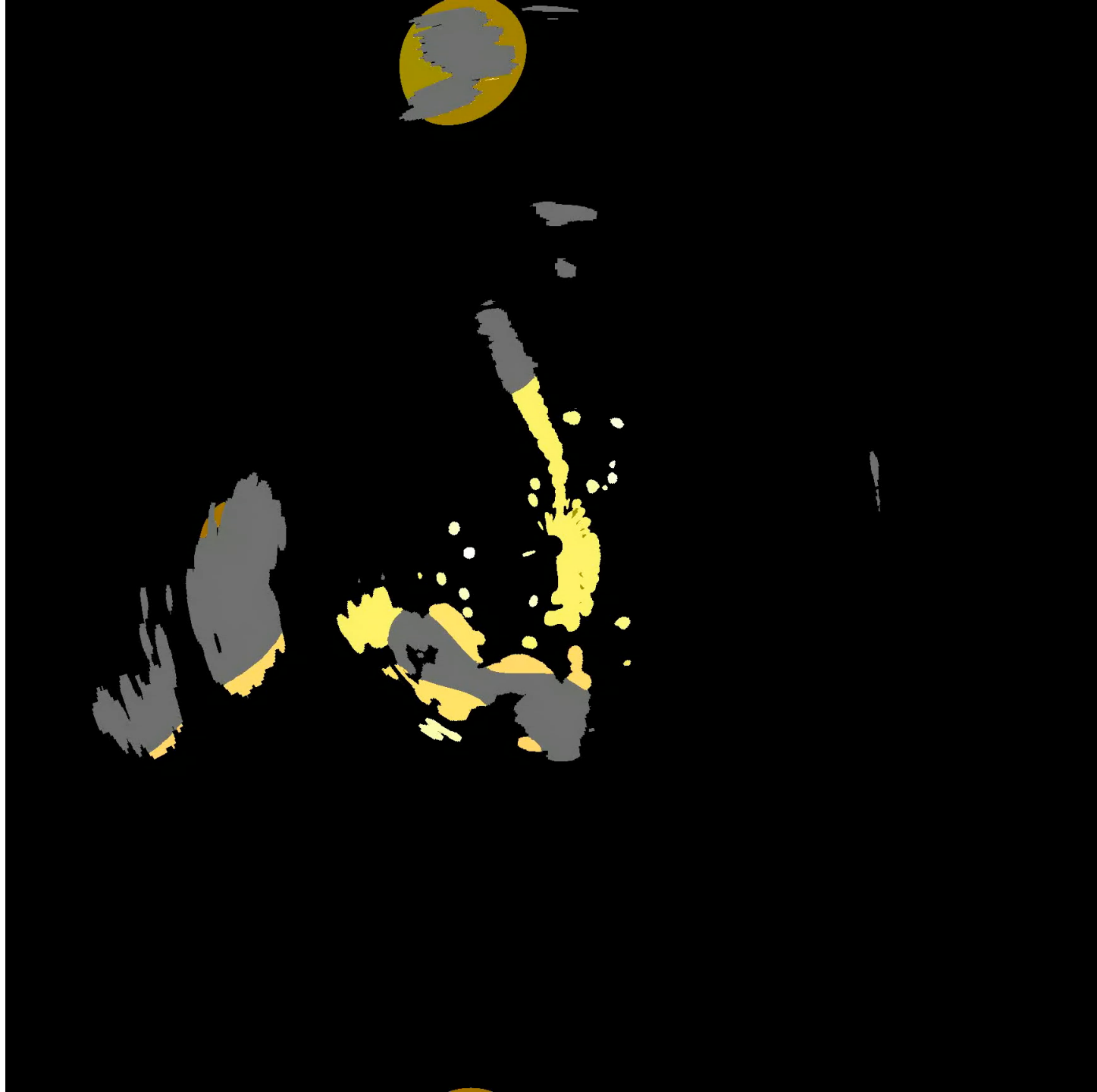
summary



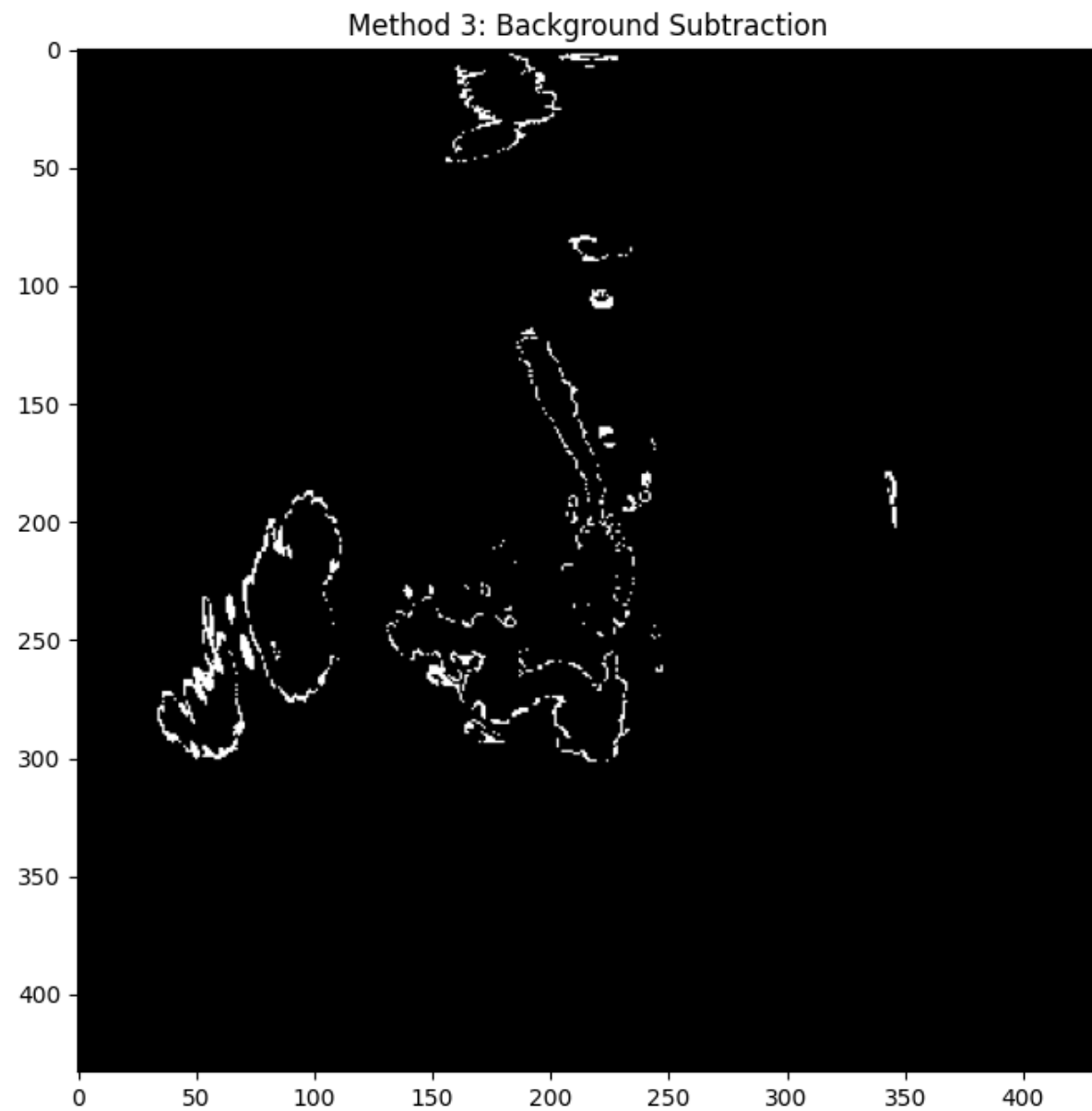
# summary



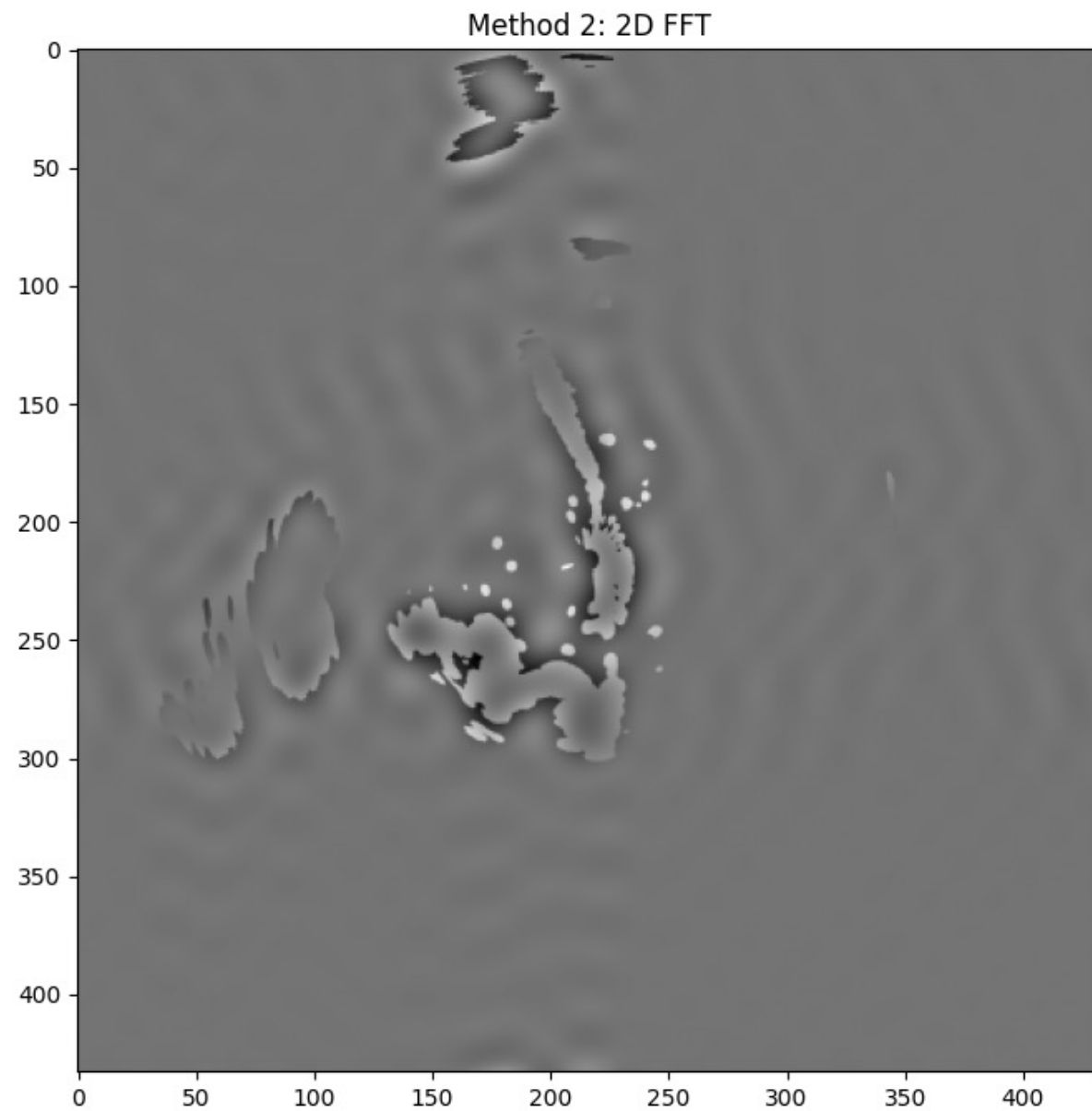
summary



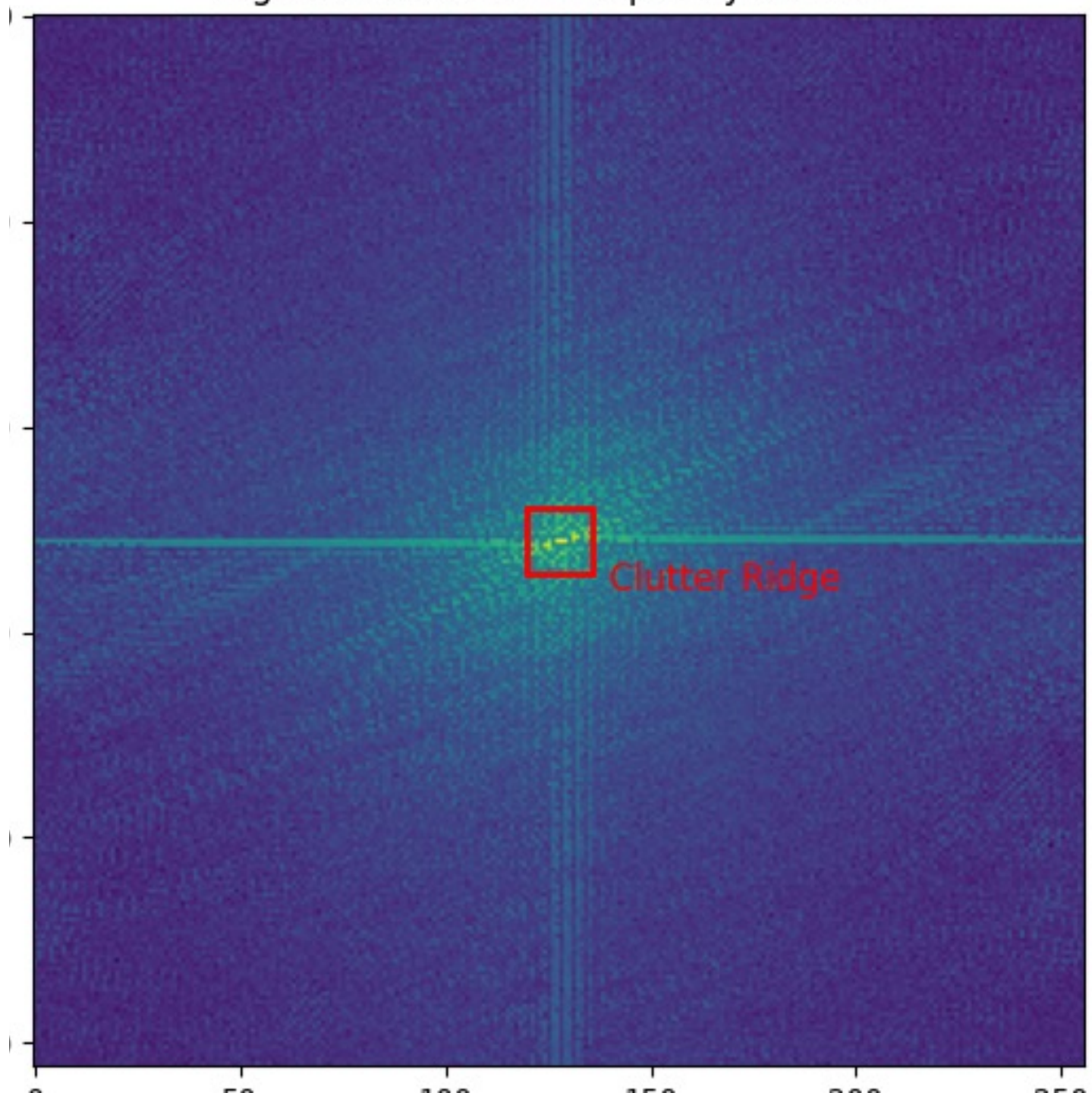
summary



summary



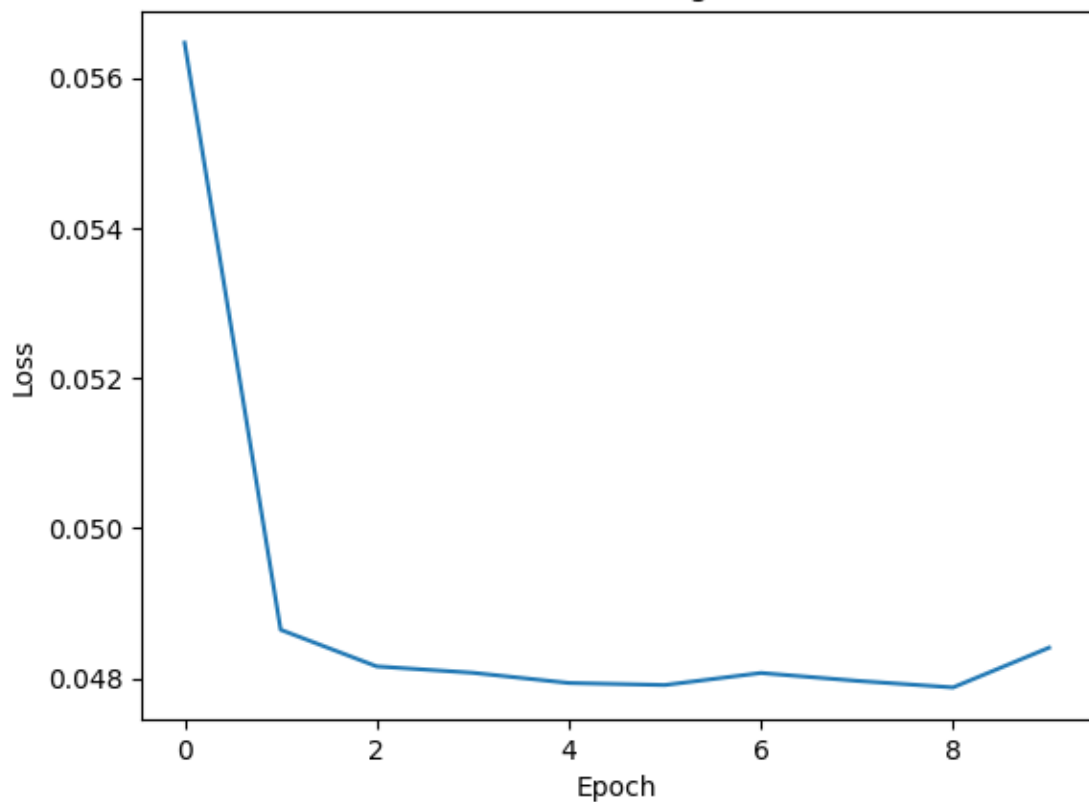
summary



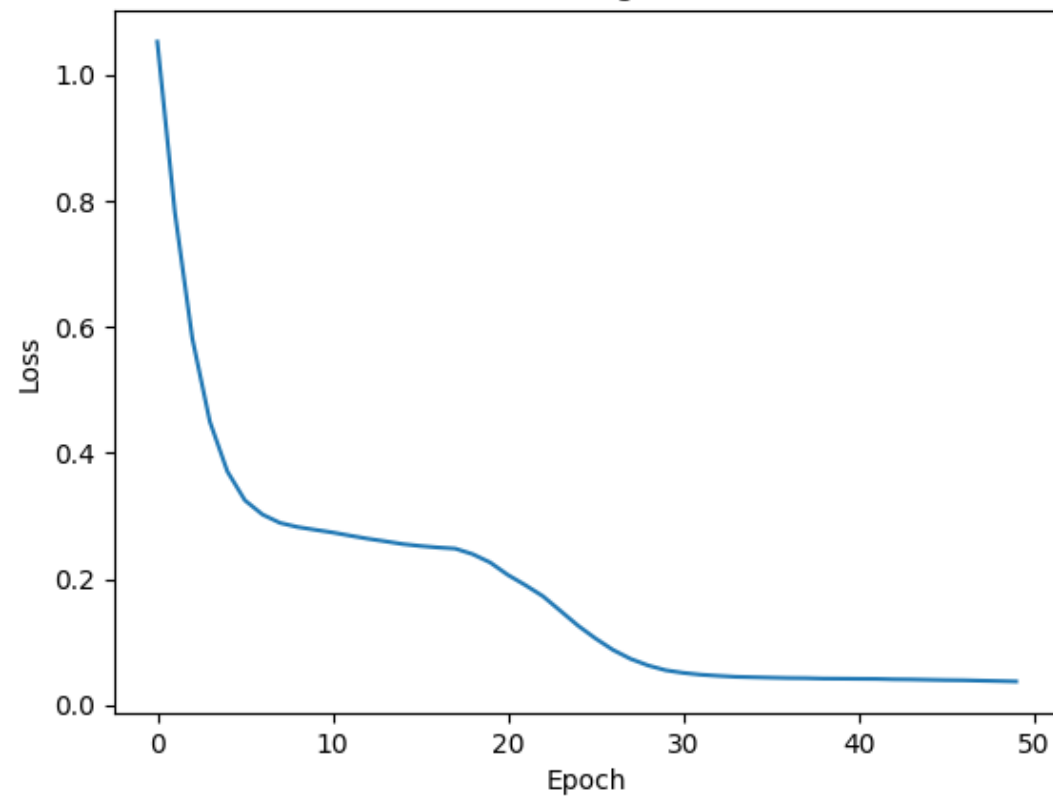
Tried building a temporal MAE at first using off-the-shelf designs.

MAE(feature extraction) → lightweight detection head(fine tuning)

MAE Pre-training Loss

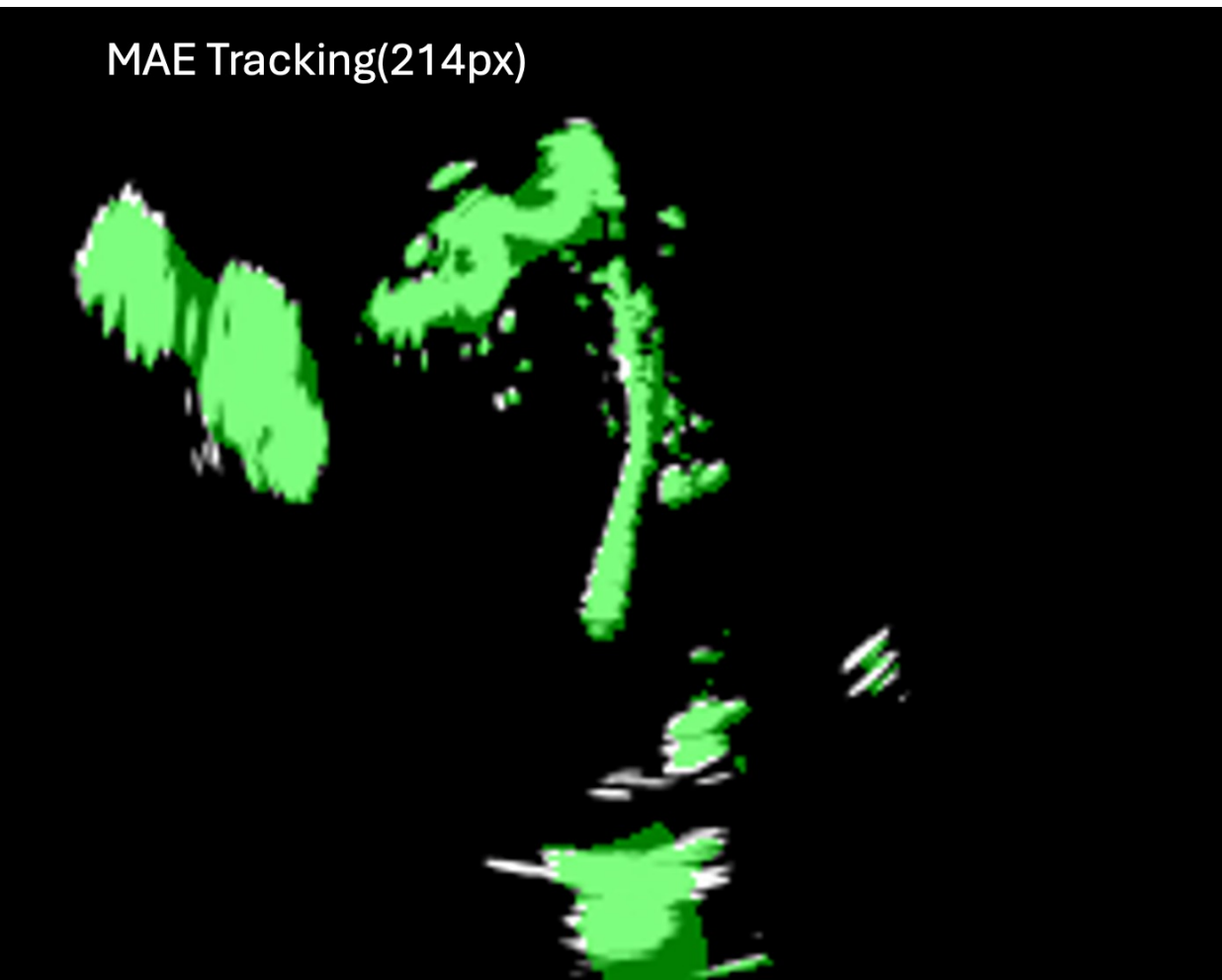


Fine-tuning Loss

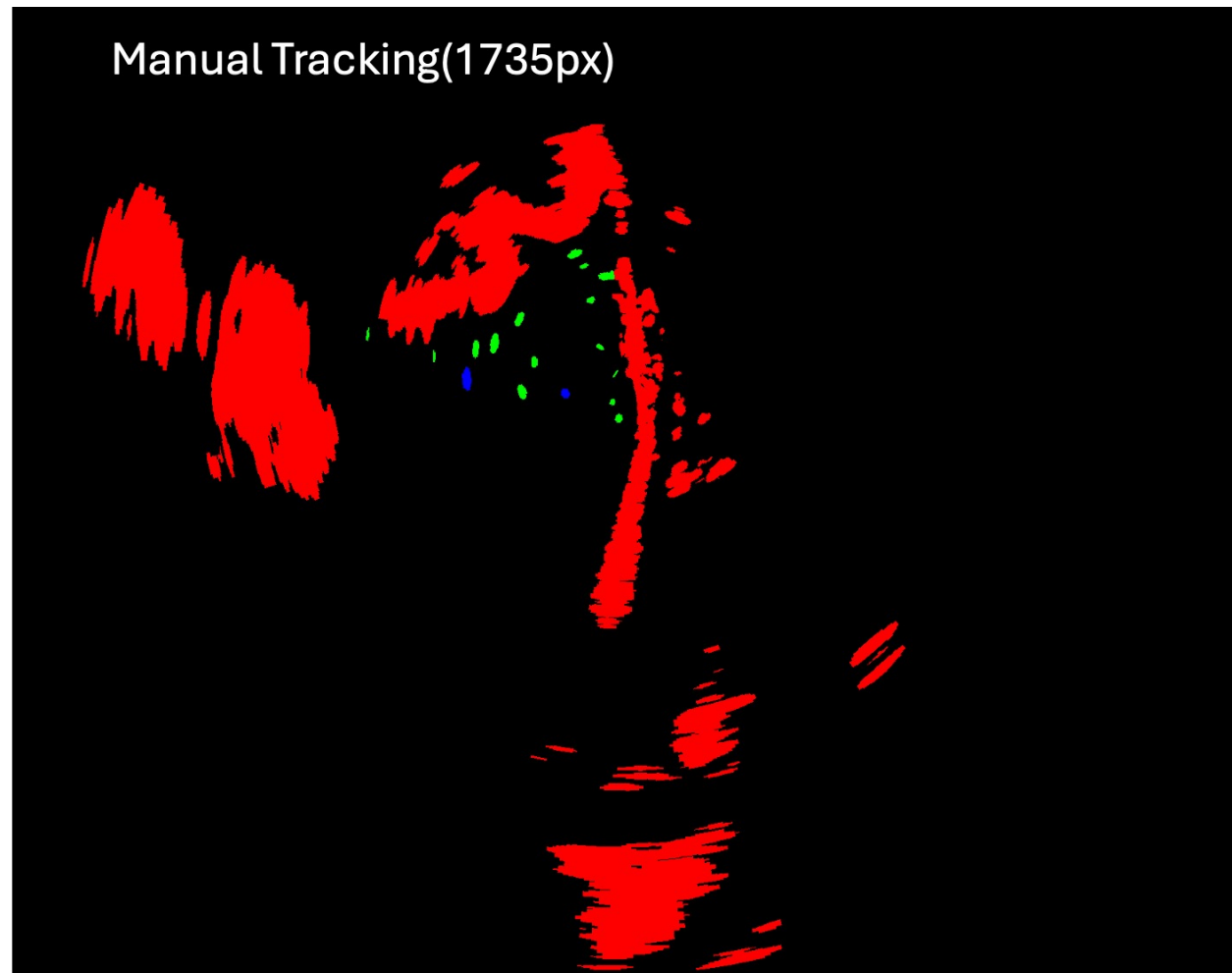


## RESULTS FROM T-MAE

MAE Tracking(214px)



Manual Tracking(1735px)



# MAE

## Implementation

| model_size | Encoder<br>Embedding Dim | Encoder<br>Depth | Encoder<br>Heads | Decoder<br>Embedding Dim | Decoder<br>Depth | Decoder<br>Heads |
|------------|--------------------------|------------------|------------------|--------------------------|------------------|------------------|
| small      | 384                      | 6                | 6                | 256                      | 4                | 8                |
| medium     | 512                      | 8                | 8                | 384                      | 6                | 12               |
| large      | 768                      | 12               | 12               | 512                      | 8                | 16               |
| xlarge     | 1024                     | 16               | 16               | 768                      | 12               | 24               |
| xxlarge    | 1280                     | 20               | 20               | 1024                     | 16               | 32               |
| huge       | 1536                     | 24               | 24               | 1280                     | 20               | 40               |
| gigantic   | 2048                     | 32               | 32               | 1536                     | 24               | 48               |

# MAE

## Implementation

| Parameter                      | Description                                                       | Default Value             |
|--------------------------------|-------------------------------------------------------------------|---------------------------|
| <code>img_size</code>          | The size of the input image (height and width).                   | 32                        |
| <code>patch_size</code>        | The size of each image patch.                                     | 4                         |
| <code>in_chans</code>          | The number of input channels (e.g., 3 for RGB).                   | 3                         |
| <code>embed_dim</code>         | The embedding dimension of the encoder.                           | 768                       |
| <code>depth</code>             | The number of Transformer blocks in the encoder.                  | 12                        |
| <code>num_heads</code>         | The number of attention heads in the encoder.                     | 12                        |
| <code>decoder_embed_dim</code> | The embedding dimension of the decoder.                           | 512                       |
| <code>decoder_depth</code>     | The number of Transformer blocks in the decoder.                  | 8                         |
| <code>decoder_num_heads</code> | The number of attention heads in the decoder.                     | 16                        |
| <code>mlp_ratio</code>         | The ratio of the MLP hidden dimension to the embedding dimension. | 4.0                       |
| <code>norm_layer</code>        | The normalization layer to use.                                   | <code>nn.LayerNorm</code> |

# MAE

## Implementation

| Parameter       | Description                                                                            | Default Value |
|-----------------|----------------------------------------------------------------------------------------|---------------|
| num_epochs      | The total number of training epochs.                                                   | 50            |
| learning_rate   | The initial learning rate for the optimizer.                                           | 1e-4          |
| mask_ratio      | The ratio of patches to mask during training.                                          | 0.75          |
| save_every      | The frequency (in epochs) at which to save model checkpoints.                          | 5             |
| visualize_every | The frequency (in epochs) at which to generate and save reconstruction visualizations. | 5             |

| Parameter    | Description                                                                  | Default for adamw |
|--------------|------------------------------------------------------------------------------|-------------------|
| weight_decay | The weight decay (L2 penalty).                                               | 0.05              |
| betas        | Coefficients used for computing running averages of gradient and its square. | (0.9, 0.999)      |
| eps          | Term added to the denominator to improve numerical stability.                | 1e-8              |

# MAE

## Implementation

| <code>scheduler_type</code> | Description                            |
|-----------------------------|----------------------------------------|
| <code>cosine</code>         | Cosine annealing scheduler (default).  |
| <code>linear</code>         | Linear decay scheduler.                |
| <code>step</code>           | Step decay scheduler.                  |
| <code>multistep</code>      | Multi-step decay scheduler.            |
| <code>exponential</code>    | Exponential decay scheduler.           |
| <code>plateau</code>        | Reduce learning rate on plateau.       |
| <code>warmup_cosine</code>  | Cosine annealing with a warmup period. |
| <code>polynomial</code>     | Polynomial decay scheduler.            |

| <code>optimizer_type</code> | Description                                  |
|-----------------------------|----------------------------------------------|
| <code>adamw</code>          | AdamW optimizer (default).                   |
| <code>adam</code>           | Adam optimizer.                              |
| <code>sgd</code>            | Stochastic Gradient Descent (SGD) optimizer. |
| <code>rmsprop</code>        | RMSprop optimizer.                           |

# MAE

## Implementation

### MAE ARCHITECTURE DEEP DIVE

#### ENCODER (ASYMMETRIC)

- **Input:** Only visible patches (~25%)
- **Architecture:** Vision Transformer
- **Purpose:** Extract features from visible content
- **Efficiency:** Processes only 25% of tokens

```
visible_patches =  
select_visible(patches) encoded =  
transformer_encoder(visible_patches)
```

#### DECODER (LIGHTWEIGHT)

- **Input:** Encoded patches + mask tokens
- **Architecture:** Smaller Transformer
- **Purpose:** Reconstruct masked patches
- **Output:** Pixel predictions for masked areas

```
full_sequence = encoded +  
mask_tokens reconstructed =  
transformer_decoder(full_sequence)
```

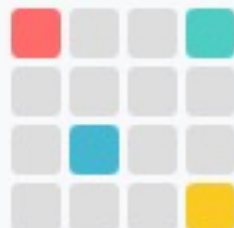
# WHAT IS A MASKED AUTOENCODER?

## MAE CORE CONCEPT

INPUT IMAGE



MASKED (75% HIDDEN)



RECONSTRUCTION



**Self-supervised learning: Learn visual representations by reconstructing randomly masked image patches**

# MAE Implementation

## MODEL ARCHITECTURE CONFIGURATIONS

### SMALL MODEL

**14M**

Parameters

---

6 Encoder Layers

4 Decoder Layers

384 Embed Dim

### MEDIUM MODEL

**36M**

Parameters

---

8 Encoder Layers

6 Decoder Layers

512 Embed Dim

### LARGE MODEL

**111M**

Parameters

---

12 Encoder Layers

8 Decoder Layers

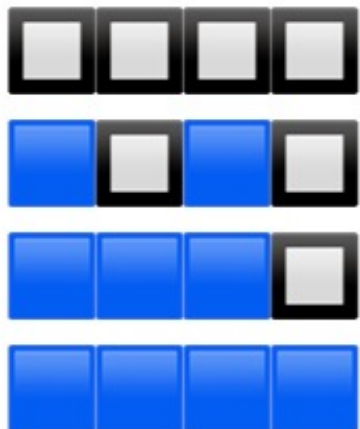
768 Embed Dim

**Each model tested with masking ratios: 35%, 50%, 60%, 75%, 90%**

# MAE Implementation

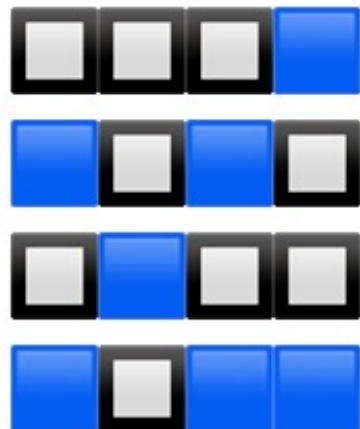
## MASKING STRATEGY IMPACT

35% MASKED



Easier reconstruction

50% MASKED



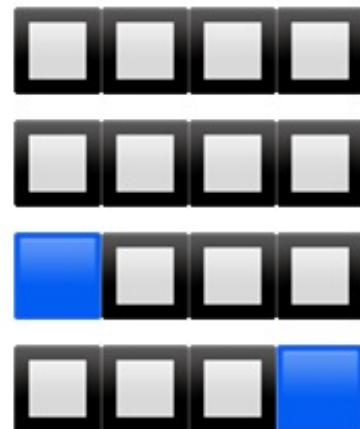
Balanced challenge

60% MASKED



More challenging

75% MASKED



Standard MAE ratio

90% MASKED



Extreme challenge

■ = Masked patches (model must predict) | ■ = Visible patches (encoder input)

# EXPERIMENT MATRIX OVERVIEW

**LARGE - 35%**

**0.189**

Test Loss

1.1h | 110.8M params

**LARGE - 50%**

**0.239**

Test Loss

1.0h | 110.8M params

**LARGE - 60%**

**0.296**

Test Loss

0.9h | 110.8M params

**LARGE - 75%**

**0.414**

Test Loss

0.7h | 110.8M params

**LARGE - 90%**

**0.709**

Test Loss

0.6h | 110.8M params

**MEDIUM - 35%**

**0.193**

Test Loss

0.4h | 36.2M params

**MEDIUM - 50%**

**0.244**

Test Loss

0.4h | 36.2M params

**MEDIUM - 60%**

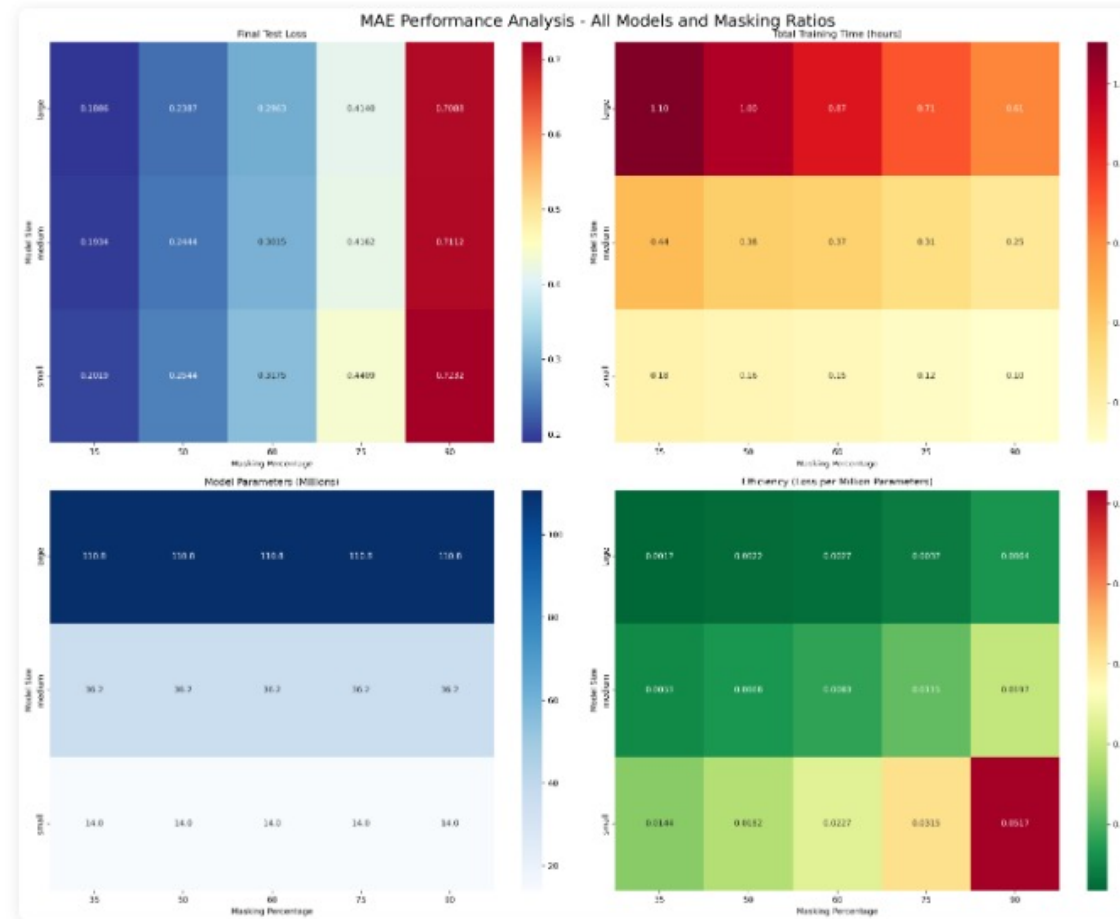
**0.301**

Test Loss

0.4h | 36.2M params

**Complete systematic exploration: 15 experiments covering all size/masking combinations**

# BEST PERFORMING CONFIGURATIONS



BEST OVERALL

**Large @ 35%**

0.189 test loss



FASTEST TRAINING

**Small @ 90%**

5.9 hours

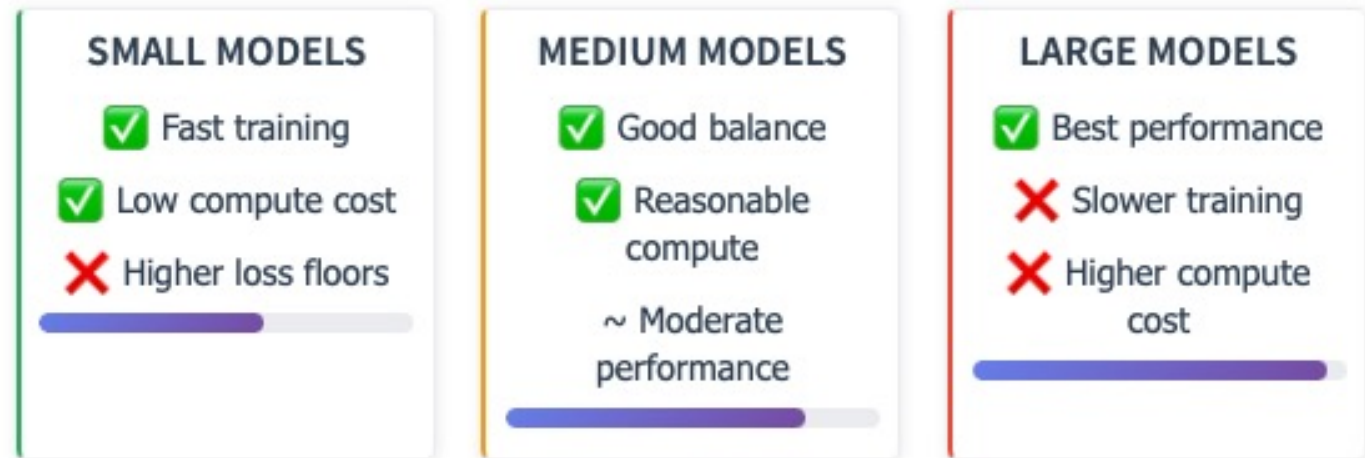


MOST EFFICIENT

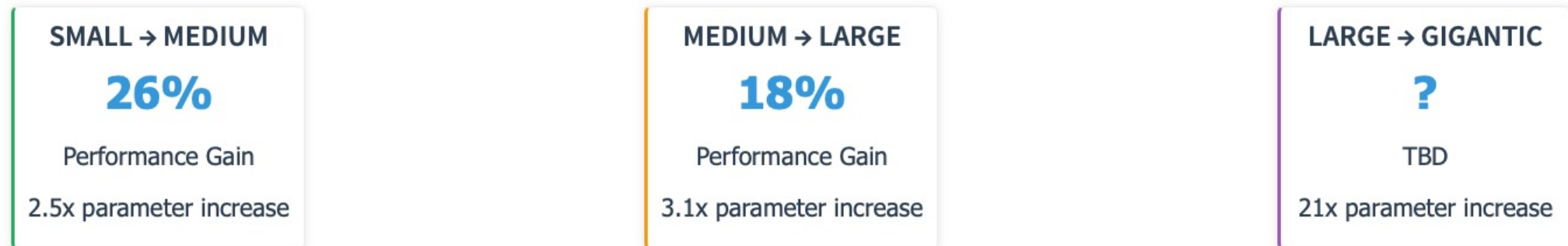
**Large @ 35%**

Best loss/param ratio

# MAE Implementation



## KEY FINDING #2: MODEL SCALING BENEFITS



Diminishing returns observed: larger models help, but gains decrease with size

# MAE Implementation

## SMALL MODEL @ 35% MASKING

### CONFIGURATION

**Encoder:** 6 layers

**Decoder:** 4 layers

**Embed Dim:** 384

**Parameters:** 14.0M

### RESULTS

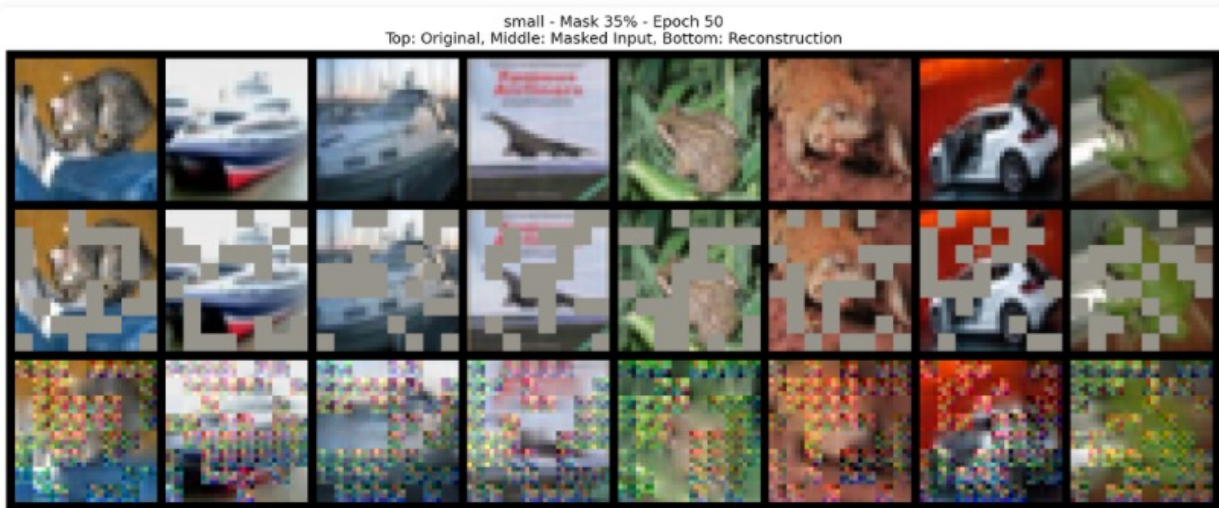
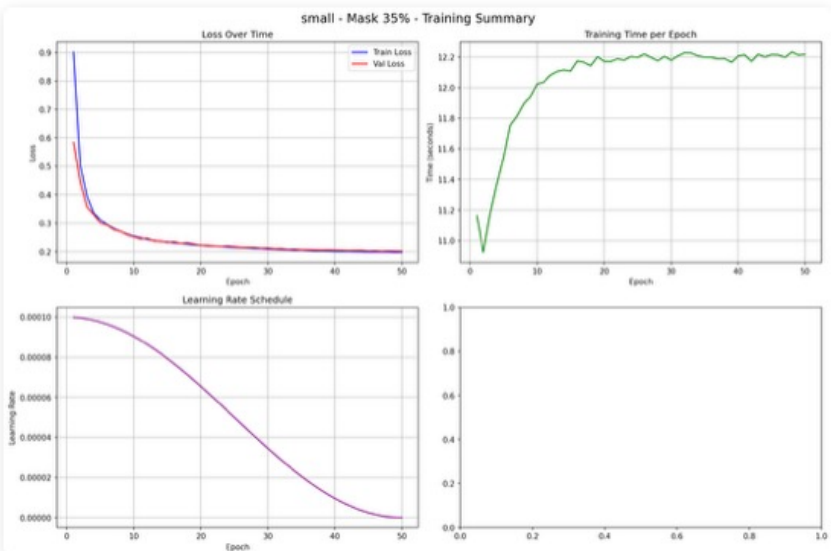
**Test Loss:** 0.202

**Best Val Loss:** 0.202

**Training Time:** 0.2h

**Avg Epoch Time:** 13.3s

### TRAINING PROGRESS & FINAL RESULTS



# MAE Implementation

## SMALL MODEL @ 75% MASKING

### CONFIGURATION

**Encoder:** 6 layers

**Decoder:** 4 layers

**Embed Dim:** 384

**Parameters:** 14.0M

### RESULTS

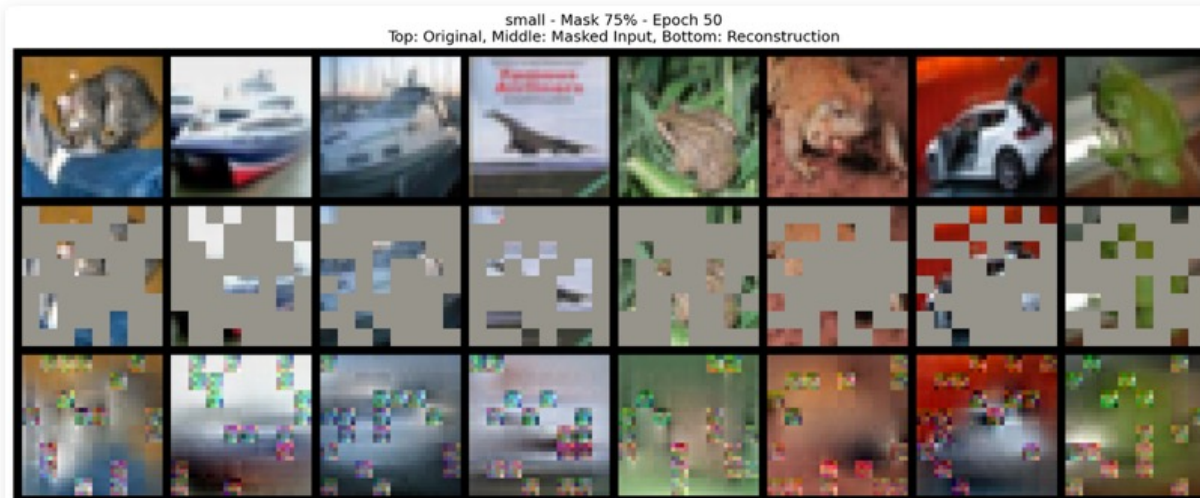
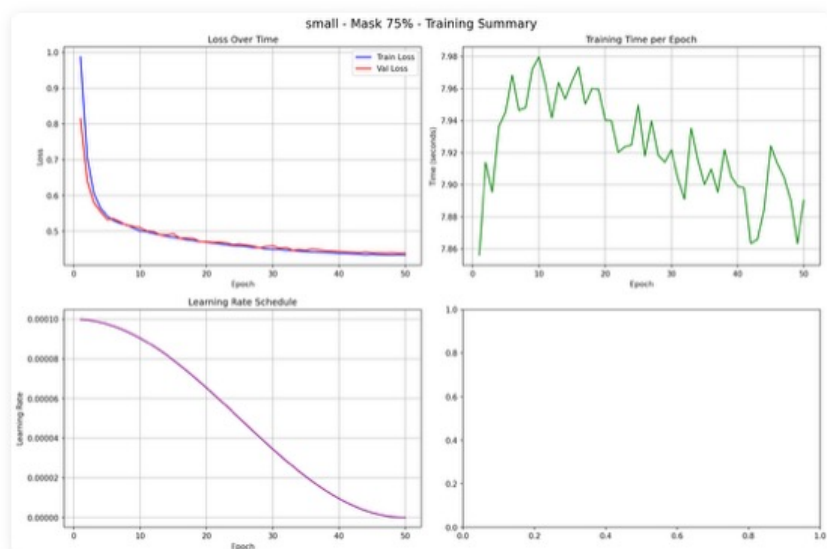
**Test Loss:** 0.441

**Best Val Loss:** 0.439

**Training Time:** 0.1h

**Avg Epoch Time:** 8.8s

### TRAINING PROGRESS & FINAL RESULTS



# MAE

## Implementation

### GIGANTIC MAE MODEL - 2.3 BILLION PARAMETERS

#### GIGANTIC MODEL

**2.3B**

Parameters

32 Encoder Layers

24 Decoder Layers

2048 Embed Dim

32 Attention Heads

✓ Adaptive Learning Rate Scheduling

✓ Early Stopping with Loss Plateau Detection

✓ Hybrid Reconstruction Visualization

✓ Neural Network Introspection

✓ Background Training with nohup

✓ Comprehensive Monitoring System

# MAE Implementation

## HYBRID RECONSTRUCTION INNOVATION

### STANDARD MAE OUTPUT

- Shows ALL patches as reconstructed
- Visible patches look "garbled"
- Hard to assess reconstruction quality
- Misleading visual representation

**Why do visible patches look bad?  
They're reconstructed, not copied!**

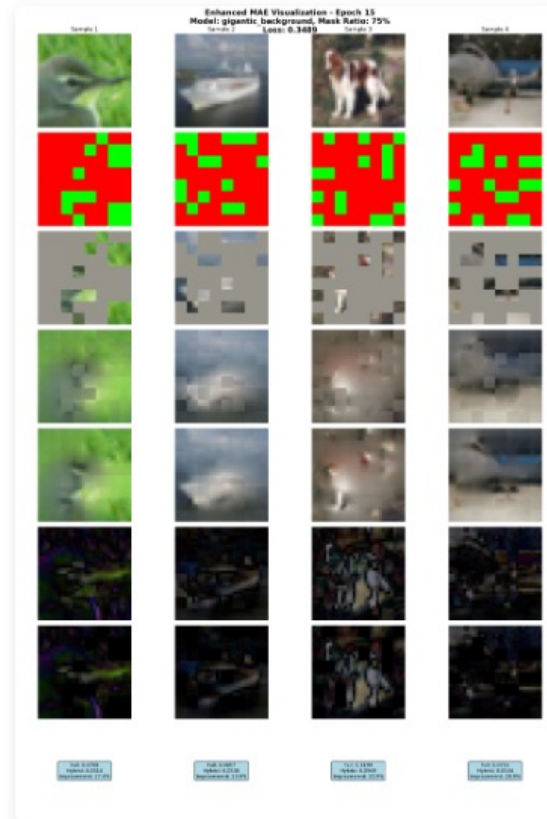
### OUR HYBRID APPROACH

- Visible patches: Original image data
- Masked patches: Model predictions
- Clear quality assessment
- Accurate representation

**True reconstruction quality:  
Only judge the predicted patches!**

# ENHANCED TRAINING VISUALIZATIONS

## GIGANTIC MODEL - ADVANCED RECONSTRUCTION ANALYSIS



### ATTENTION PATTERNS

Visualize which patches the model focuses on

### ACTIVATION DISTRIBUTIONS

Monitor neuron behavior across layers

### WEIGHT EVOLUTION

Track parameter changes during training

### 3D EMBEDDINGS

PCA visualization of learned representations

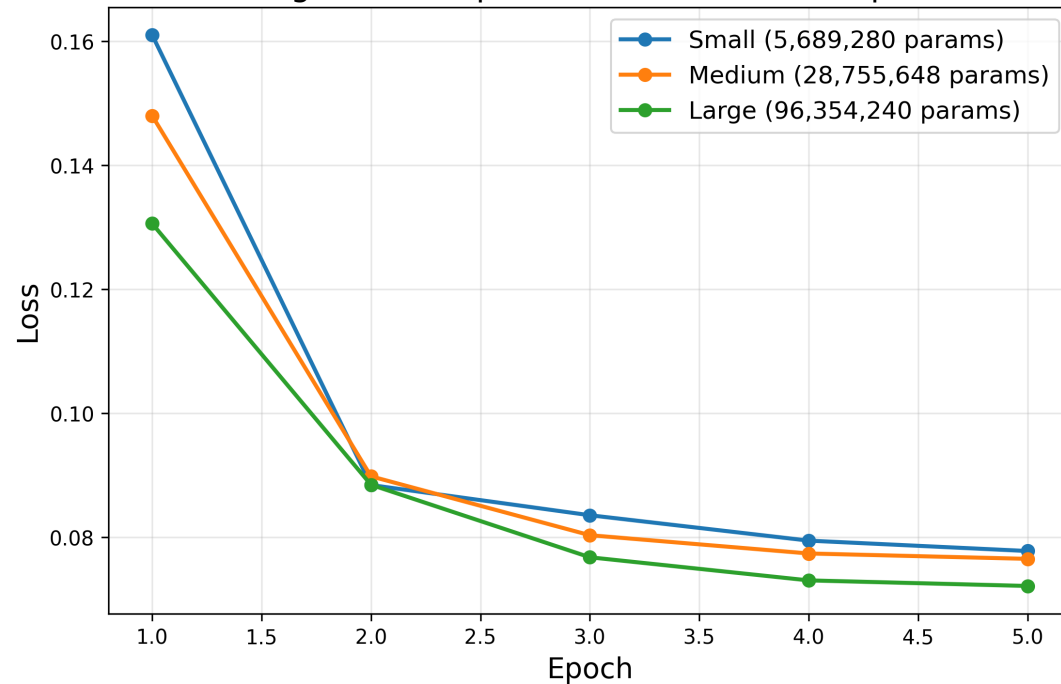
# MAE

## Implementation

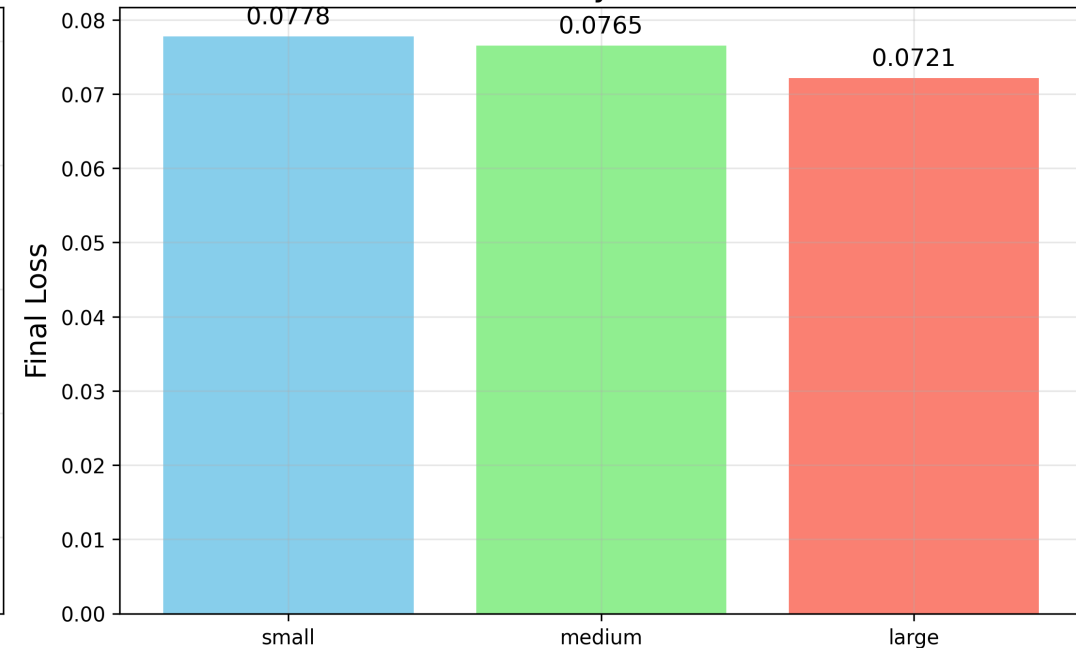
### LESSONS LEARNED & FUTURE DIRECTIONS

-  **MASKING STRATEGY MATTERS MORE THAN EXPECTED** <sup>\*sometimes</sup>  
The standard 75% masking is not optimal for all scenarios - lower ratios ~~often~~ work better
-  **SCALING LAWS ARE COMPLEX**  
Larger models help, but with diminishing returns - 2x parameters  $\neq$  2x better performance
-  **ADAPTIVE TRAINING IS ESSENTIAL**  
Learning rate scheduling and early stopping significantly improve efficiency and results
-  **VISUALIZATION DRIVES UNDERSTANDING**  
Hybrid reconstructions and neural visualizations provide crucial insights into model behavior

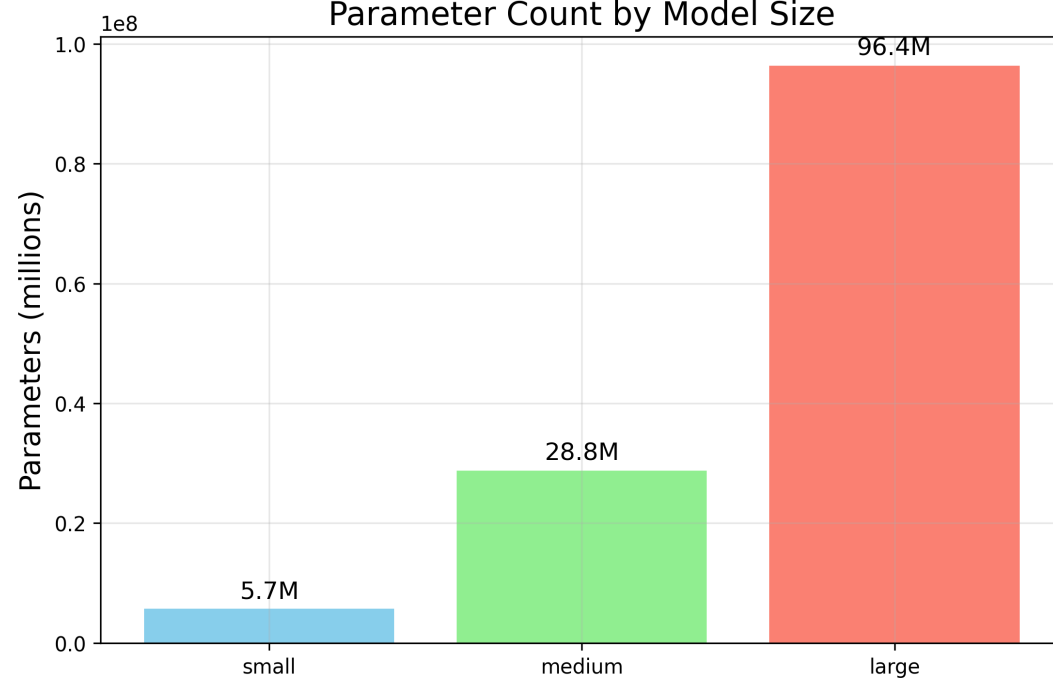
### Training Loss Comparison Across Model Capacities



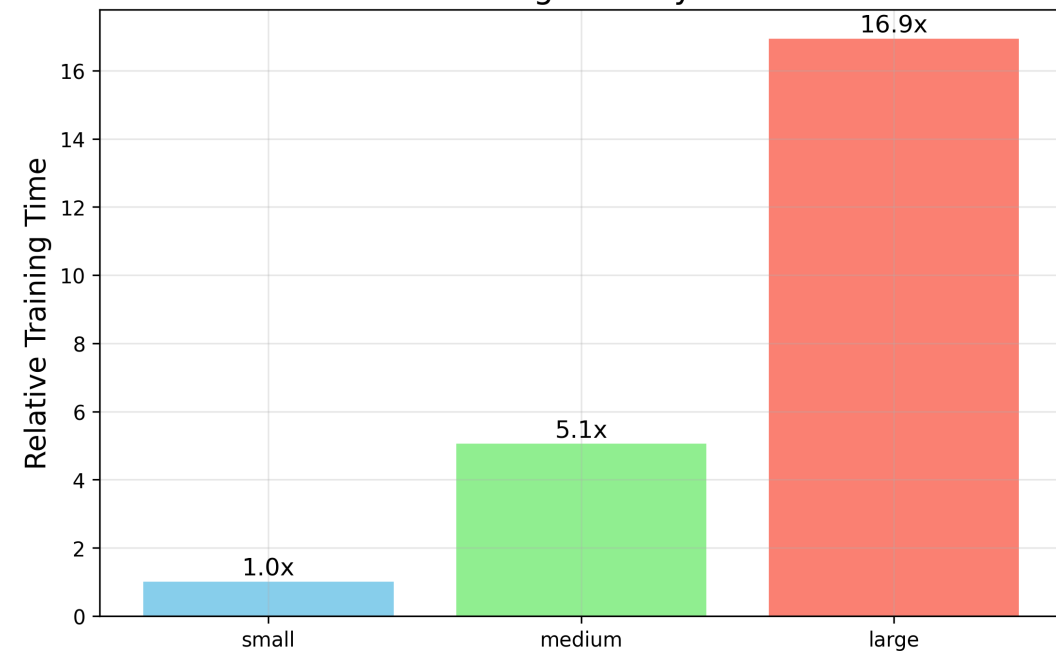
### Final Loss by Model Size



### Parameter Count by Model Size

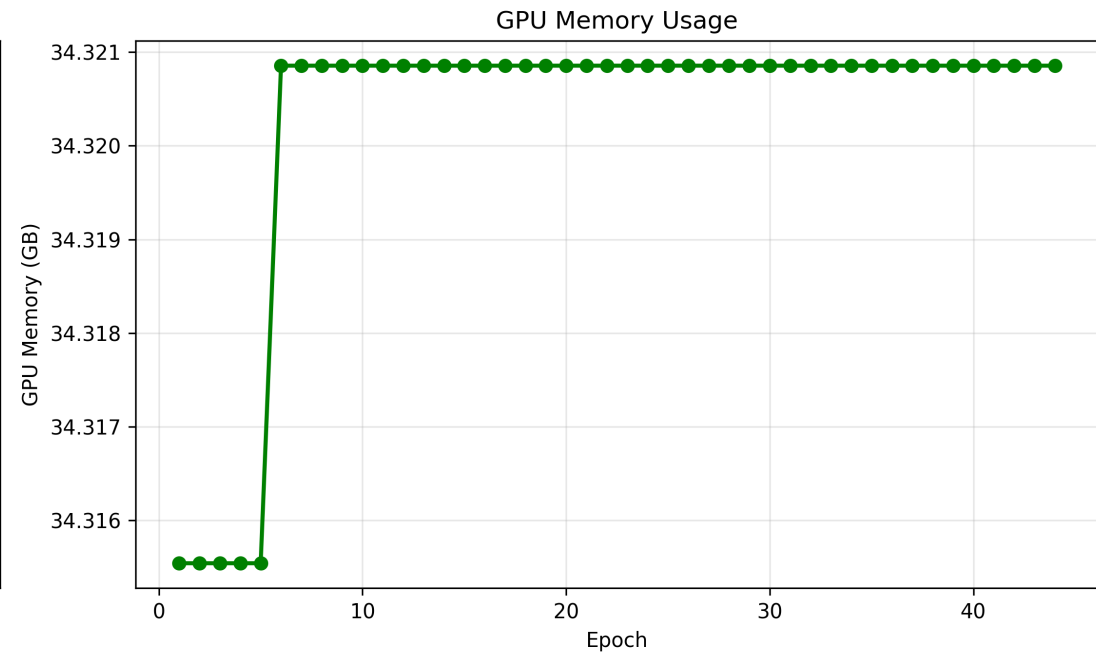
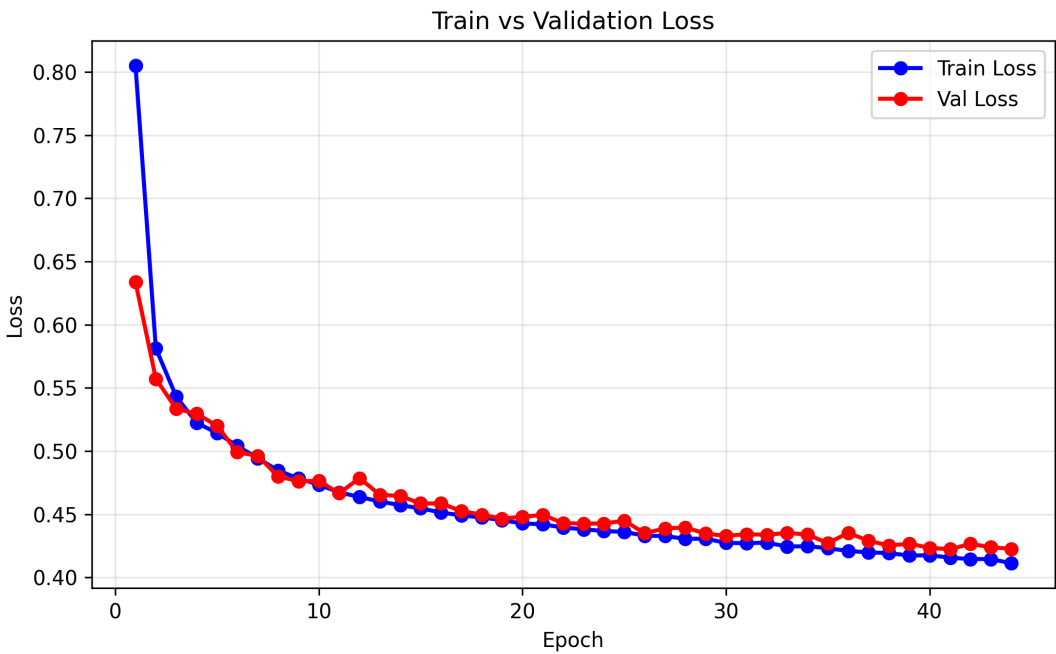
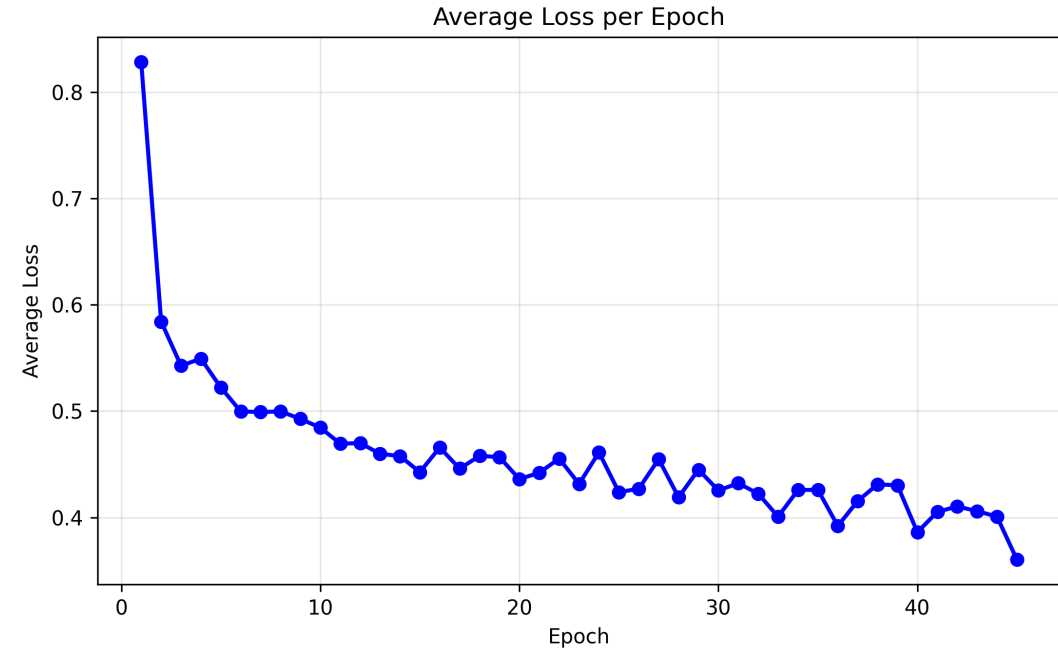
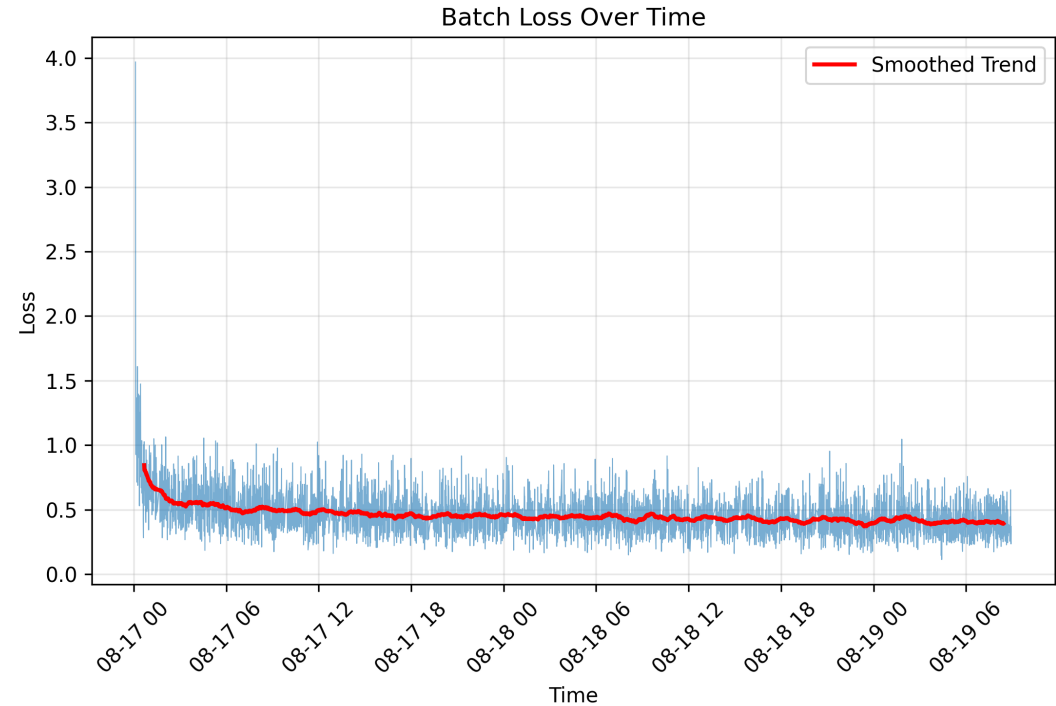


### Relative Training Time by Model Size



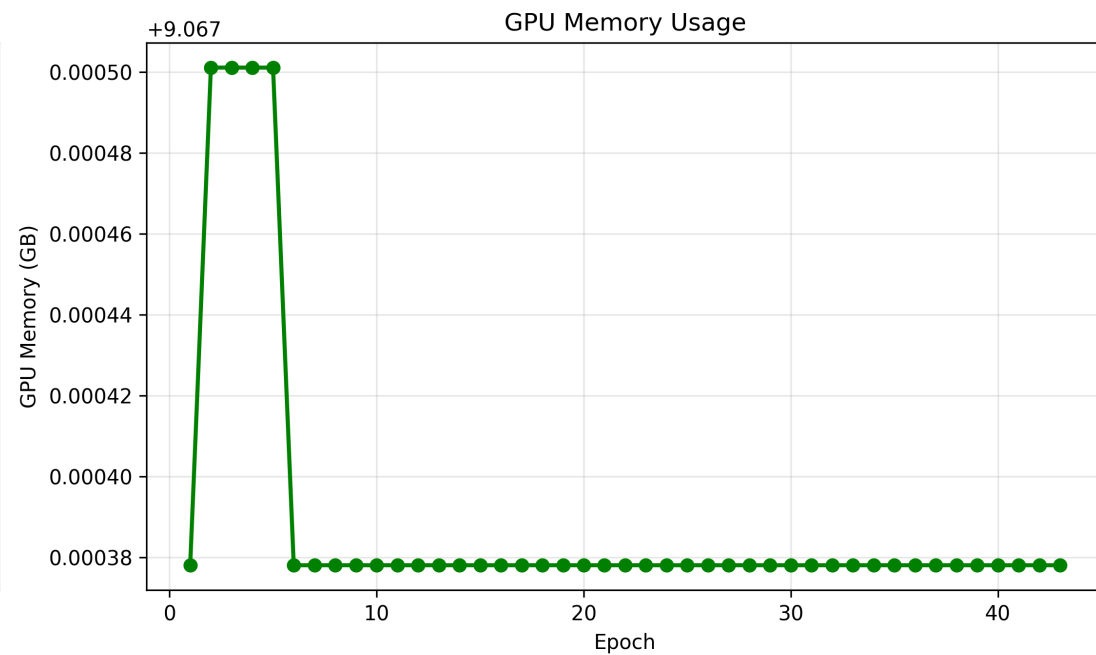
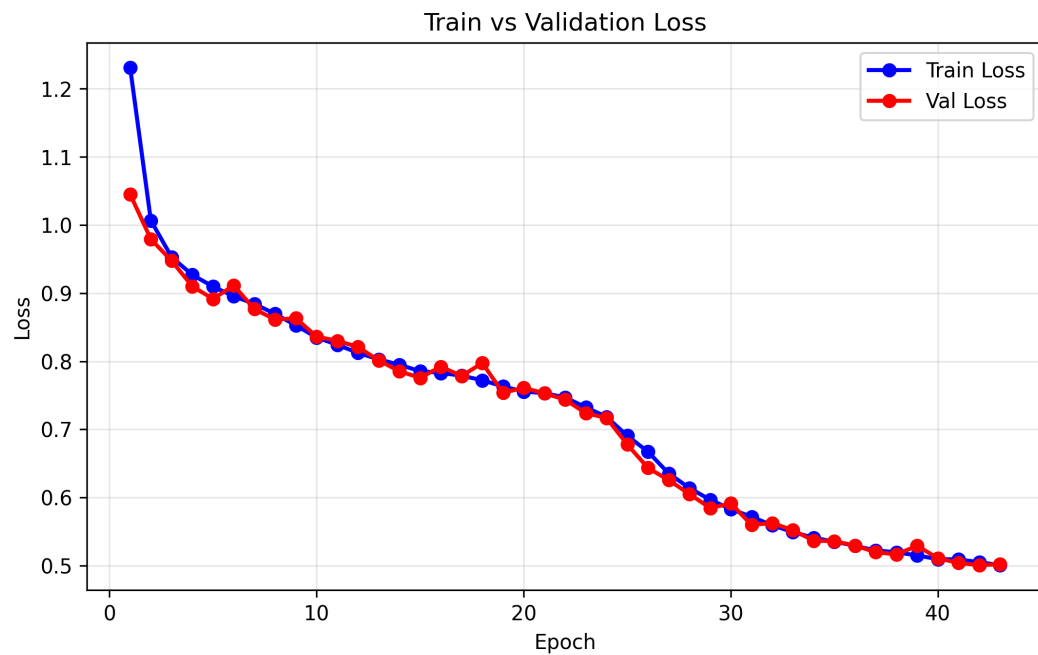
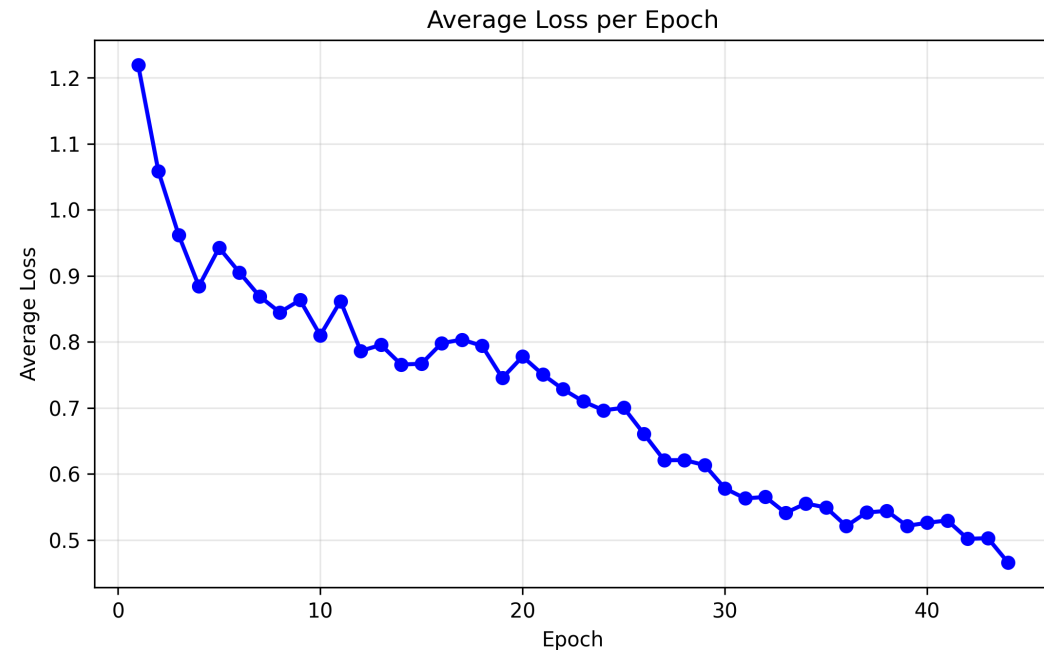
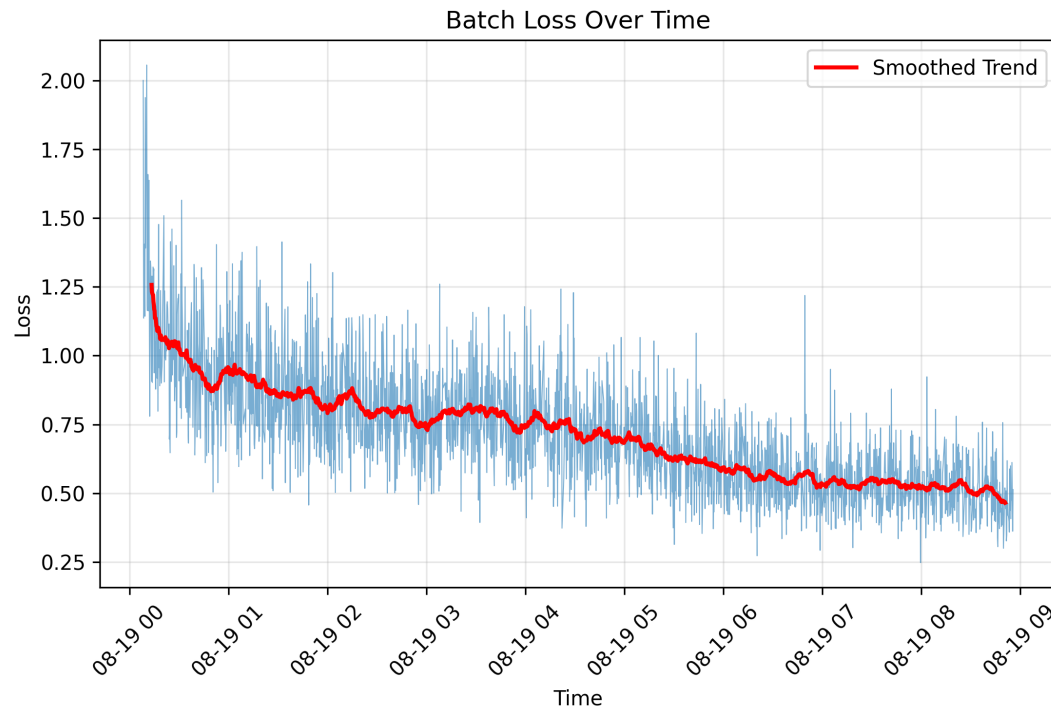
2.3B

# Training Progress - Gigantic Background Experiment



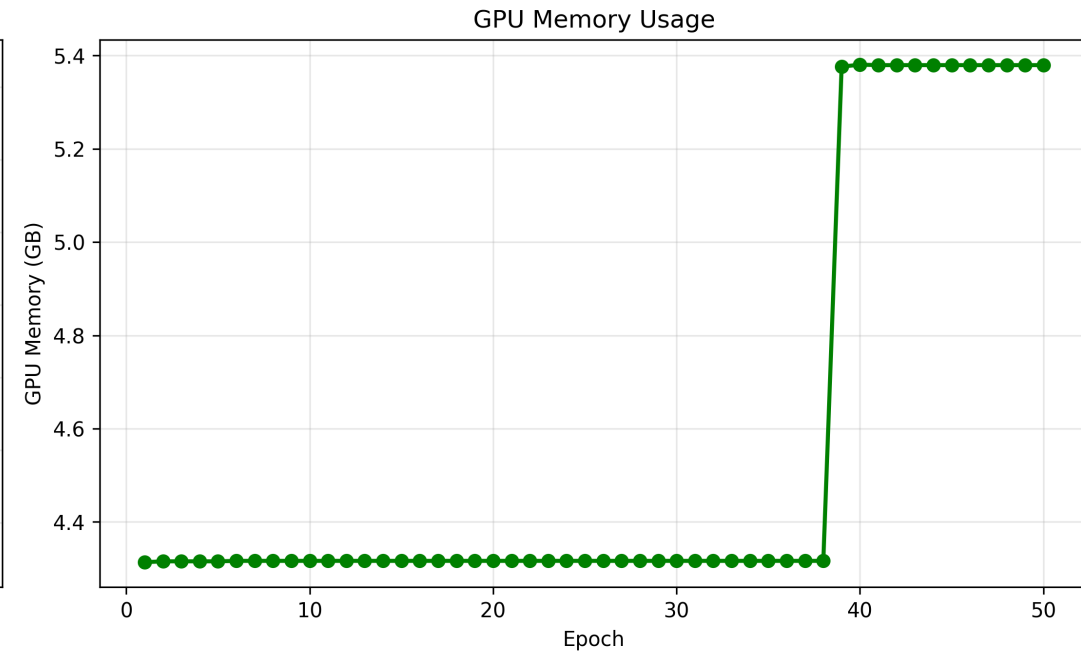
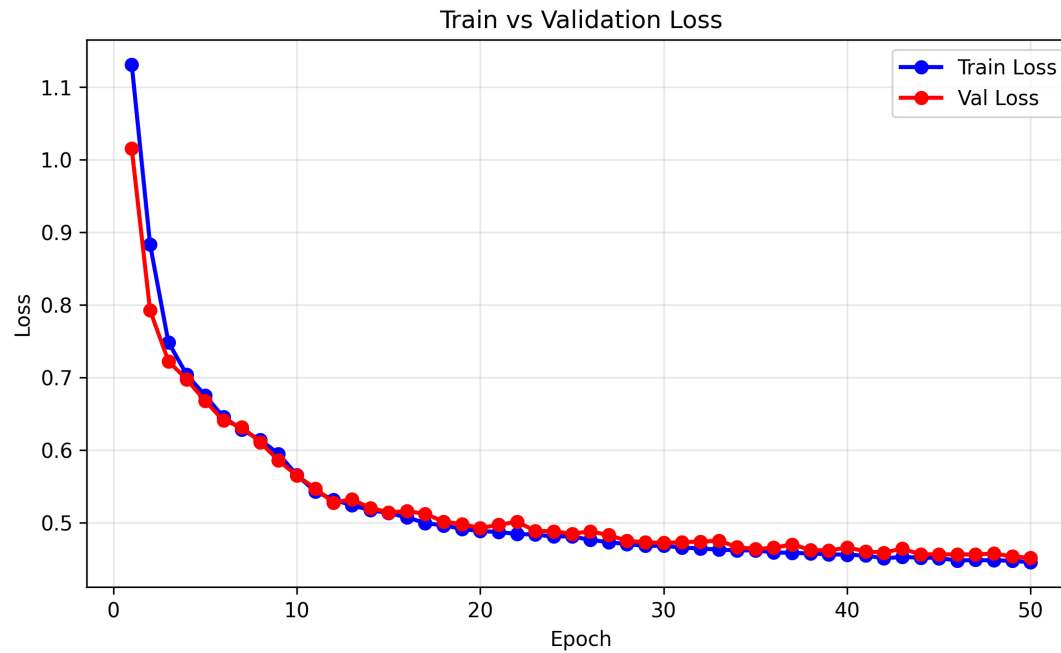
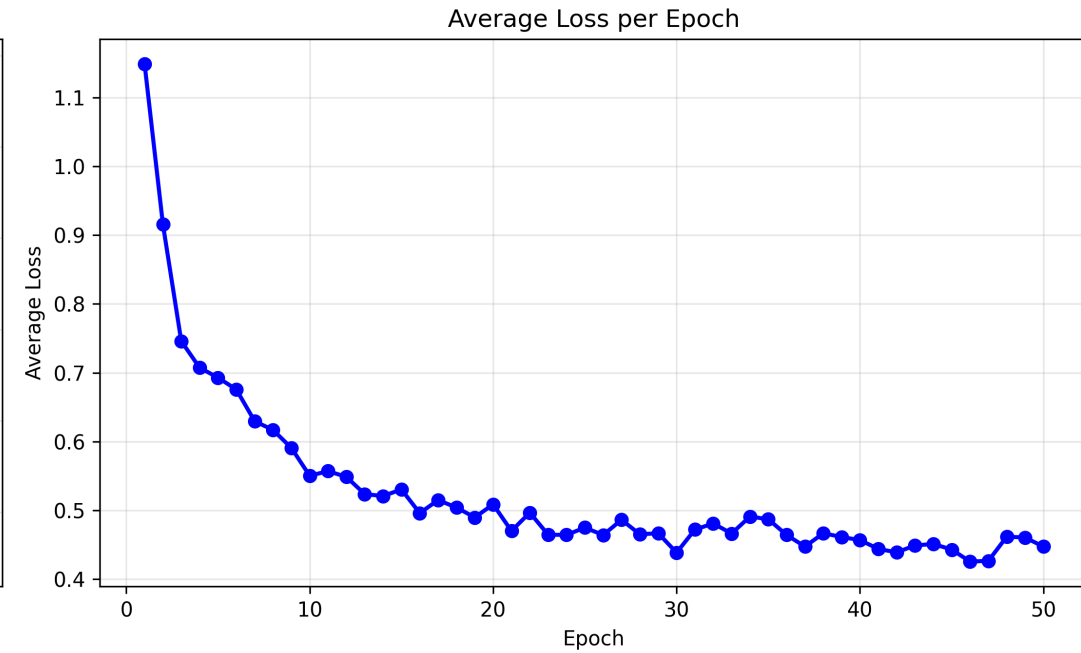
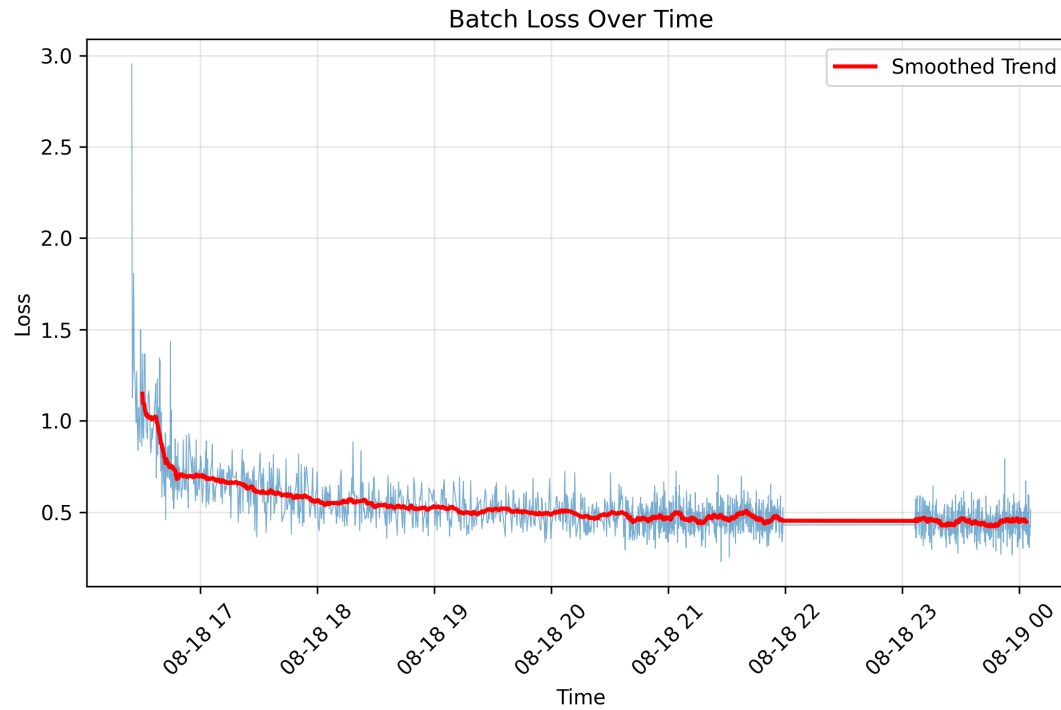
XXL

# Training Progress - Gigantic Background Experiment



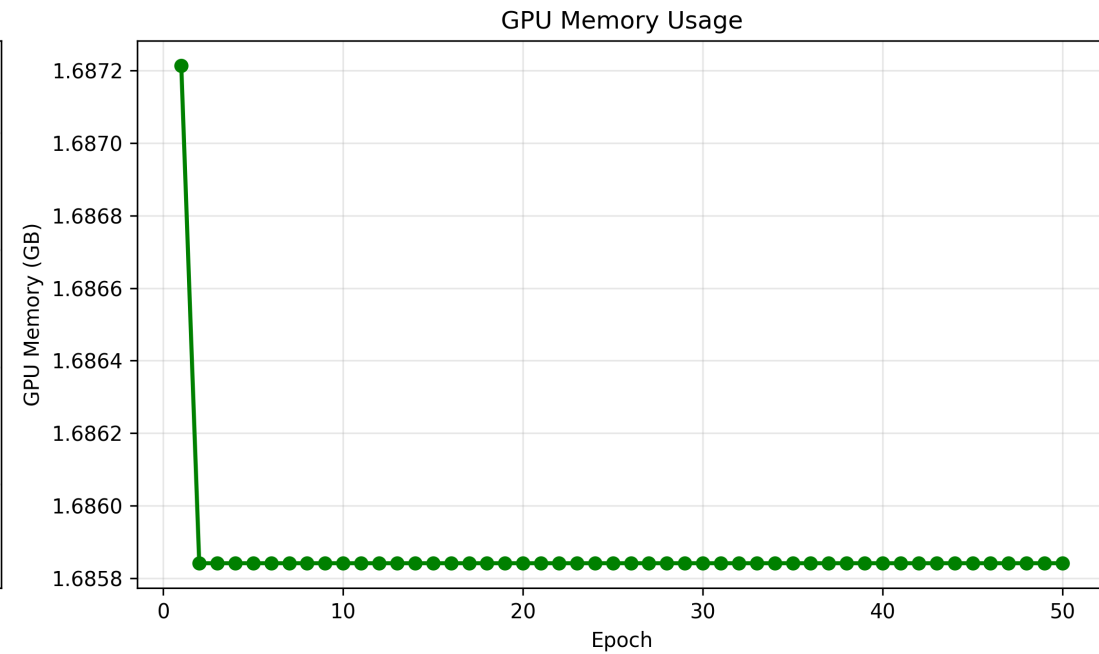
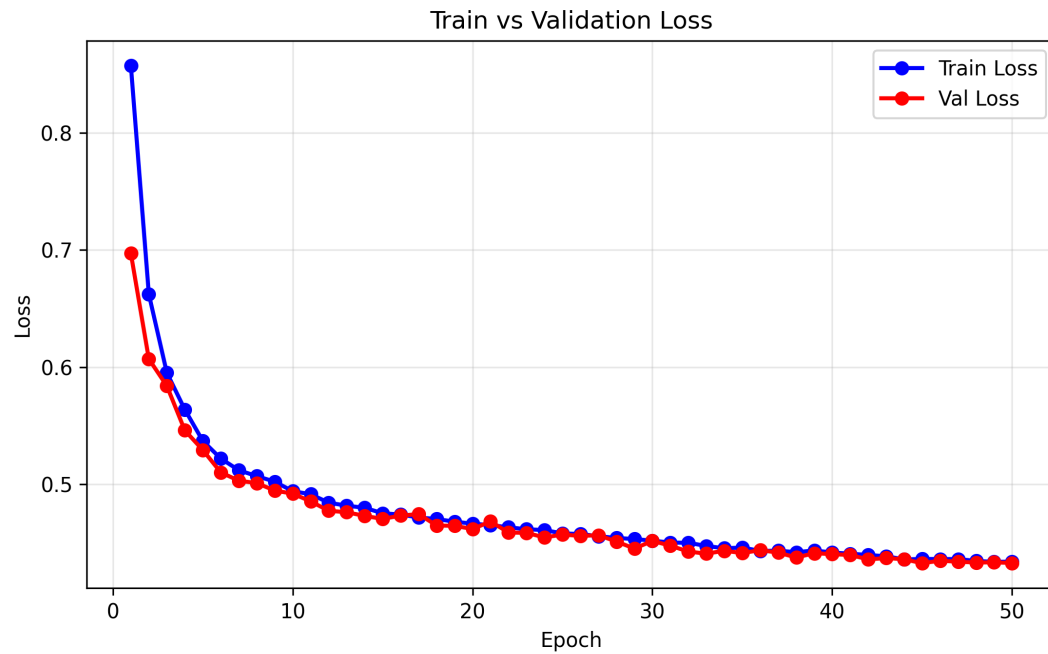
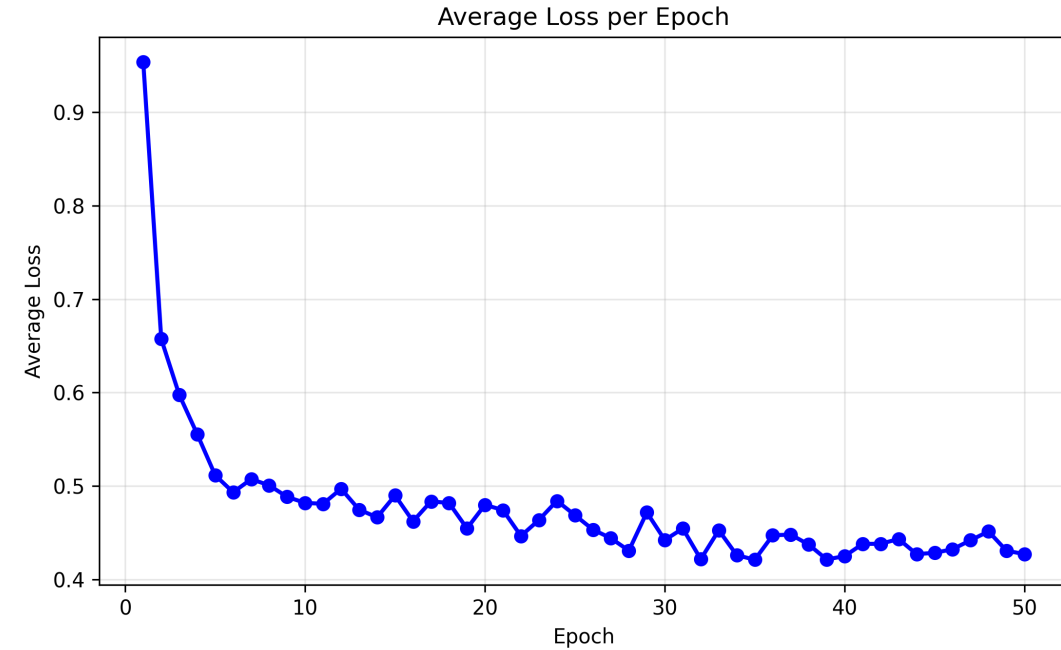
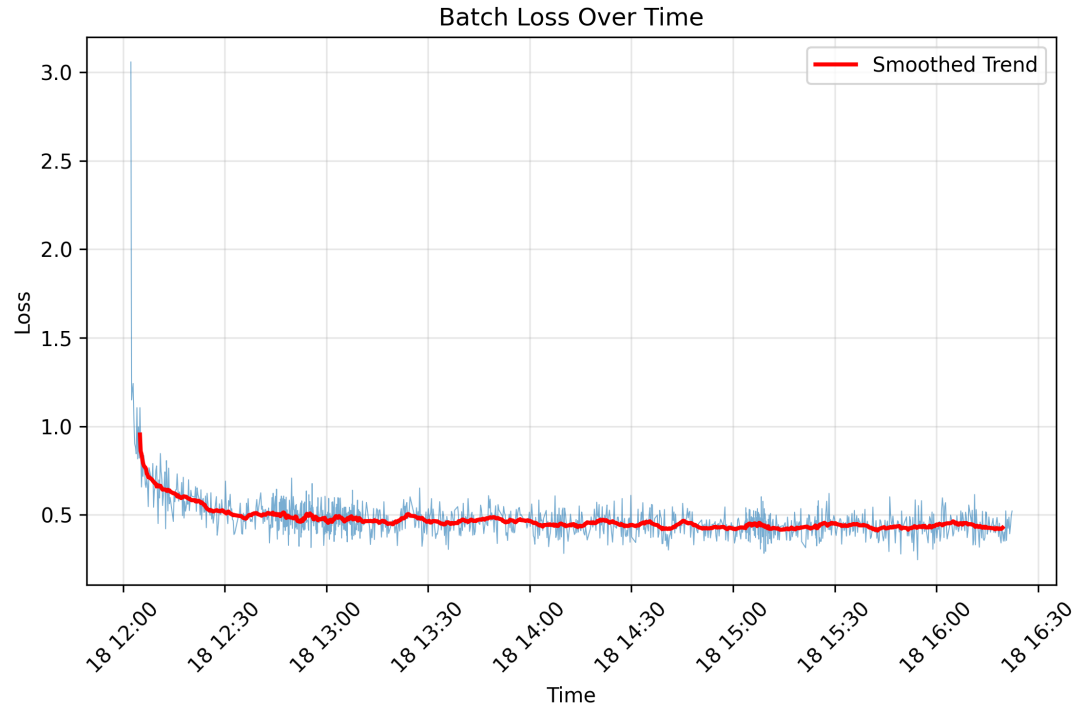
XL

# Training Progress - Gigantic Background Experiment



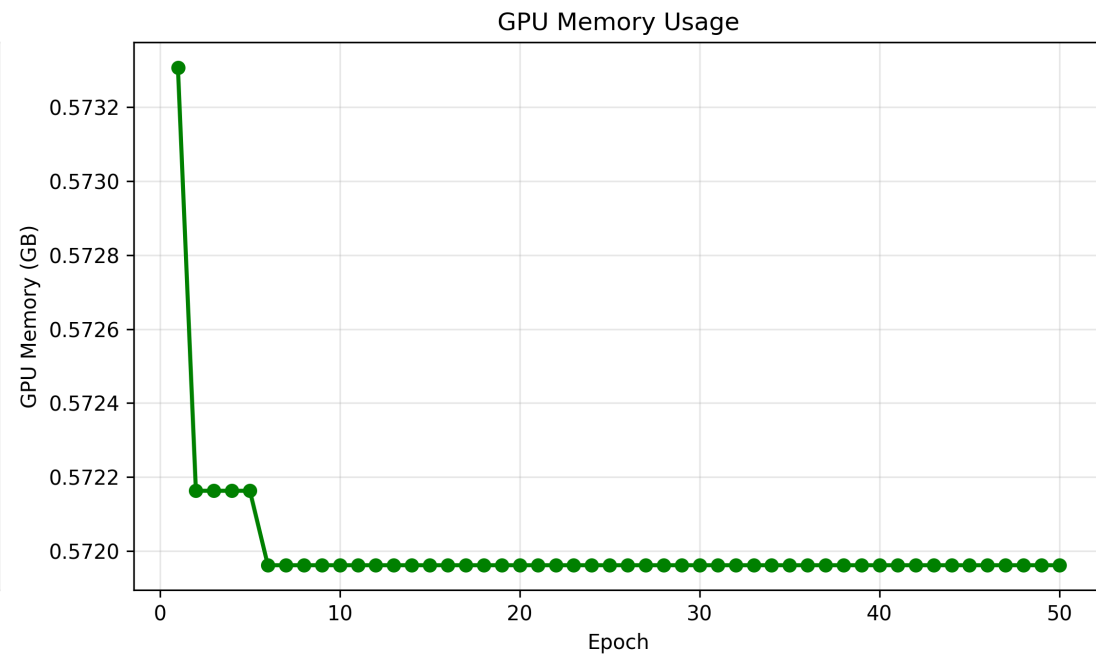
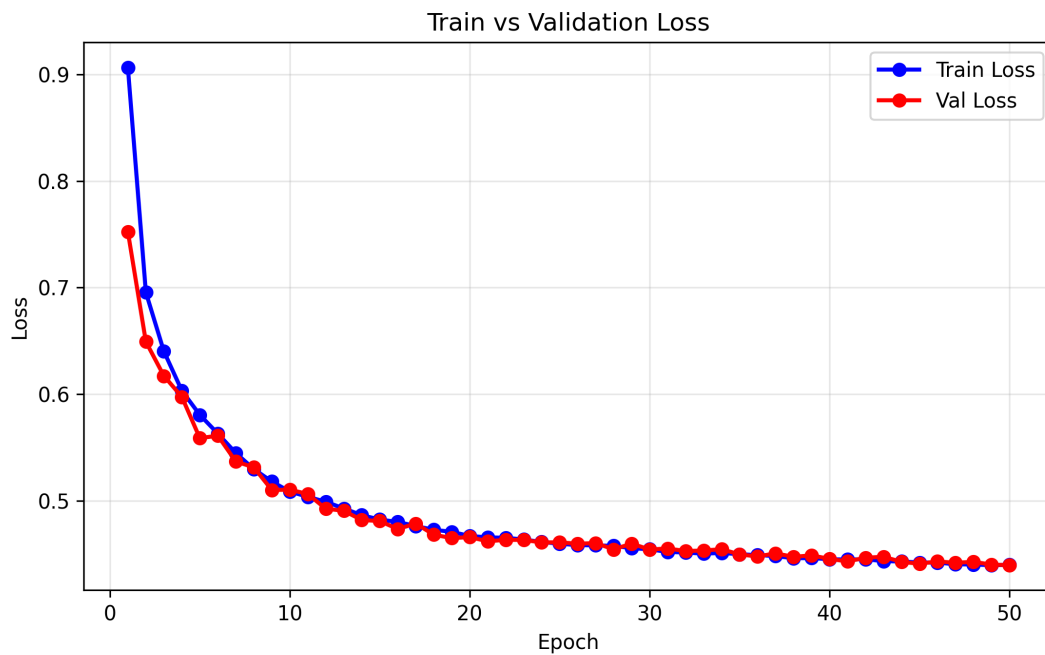
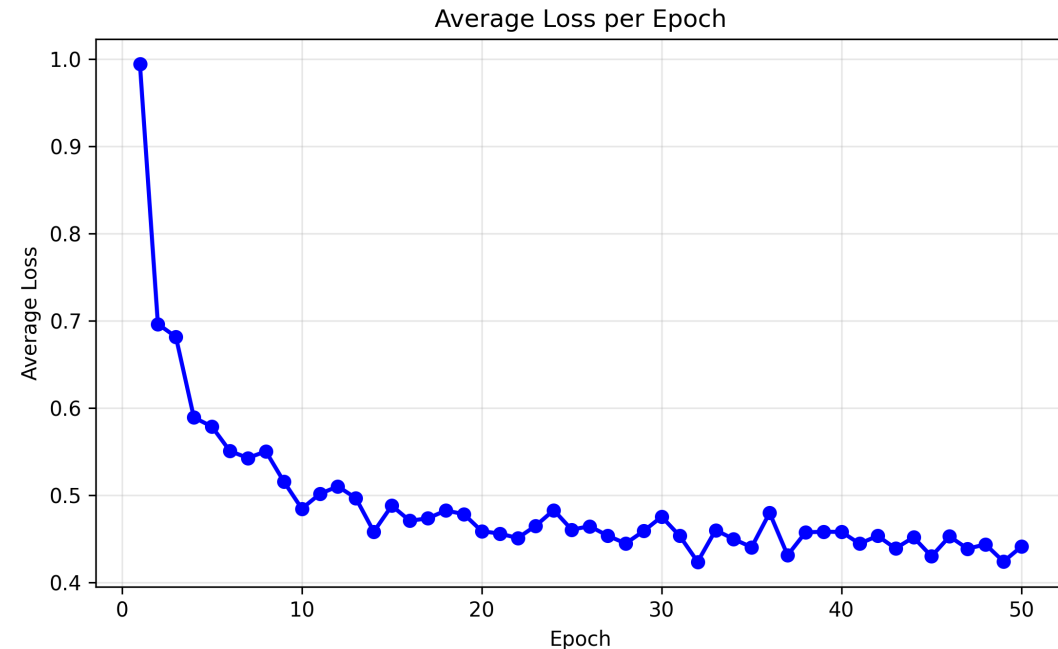
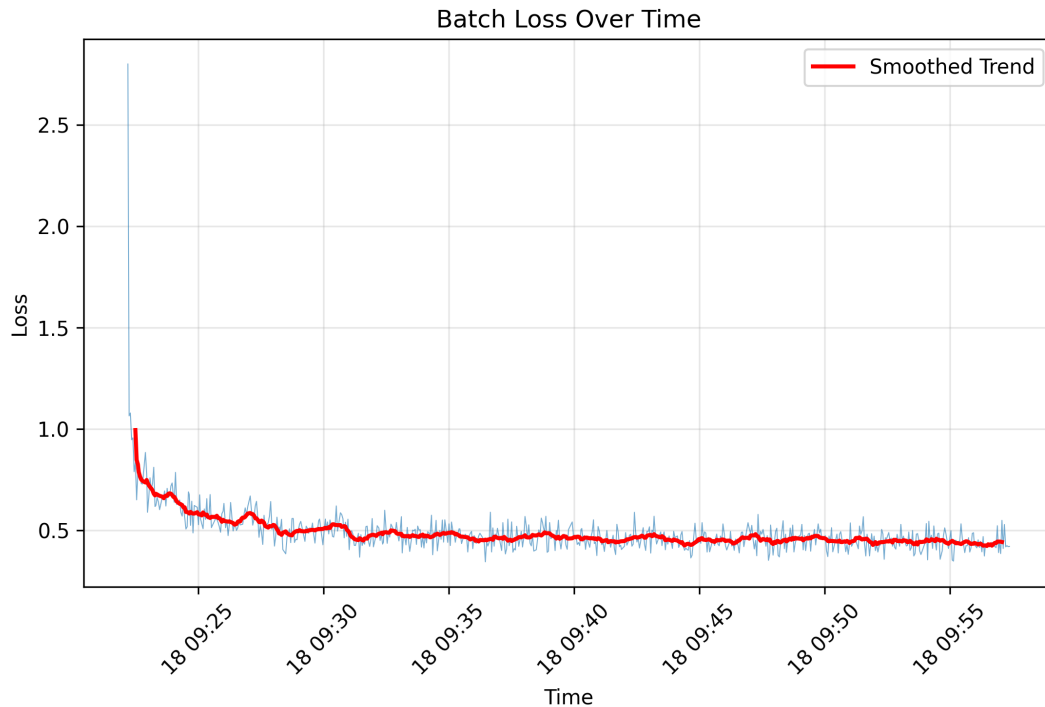
L

# Training Progress - Gigantic Background Experiment



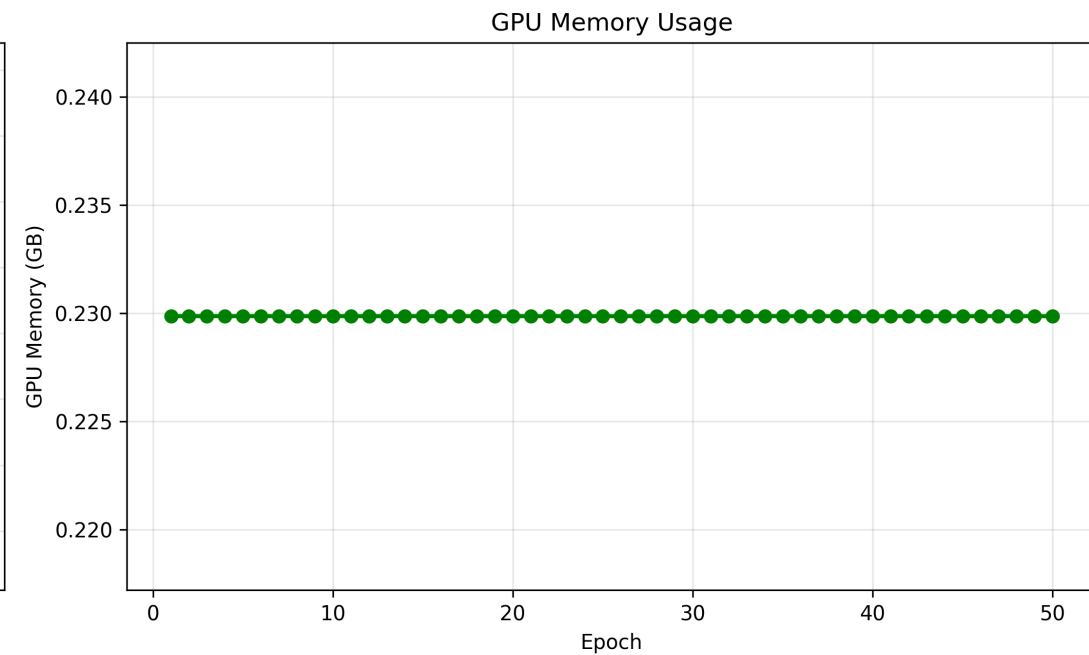
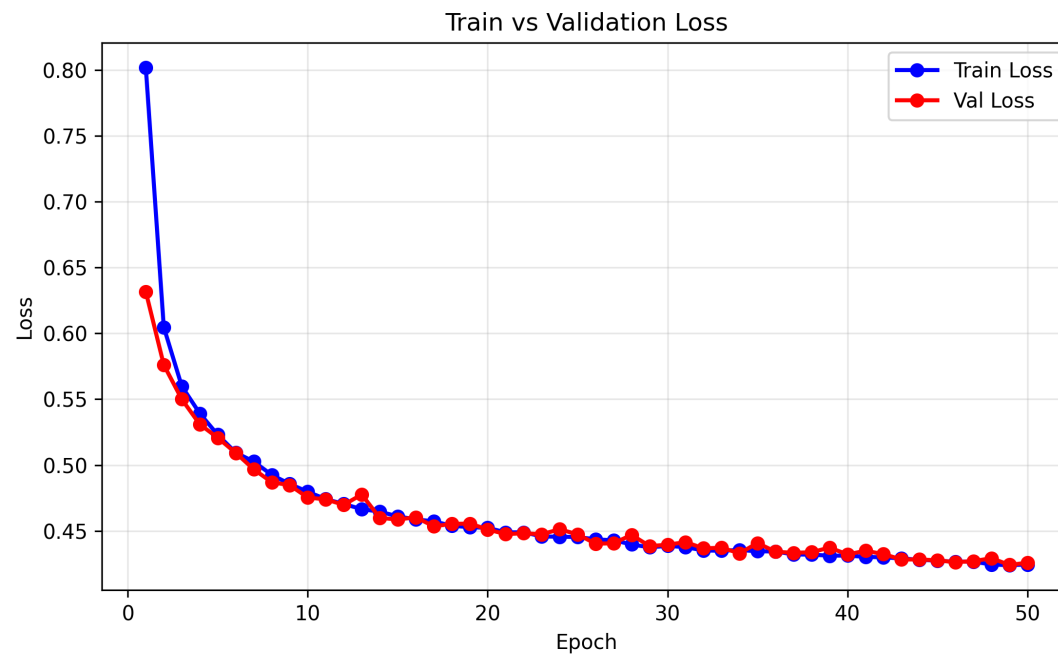
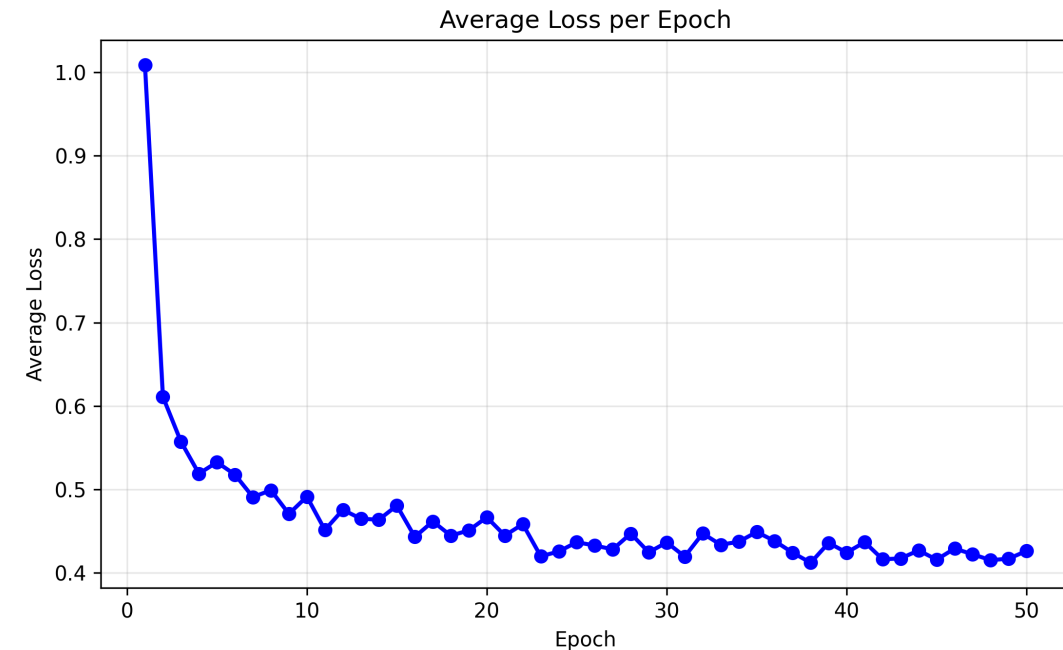
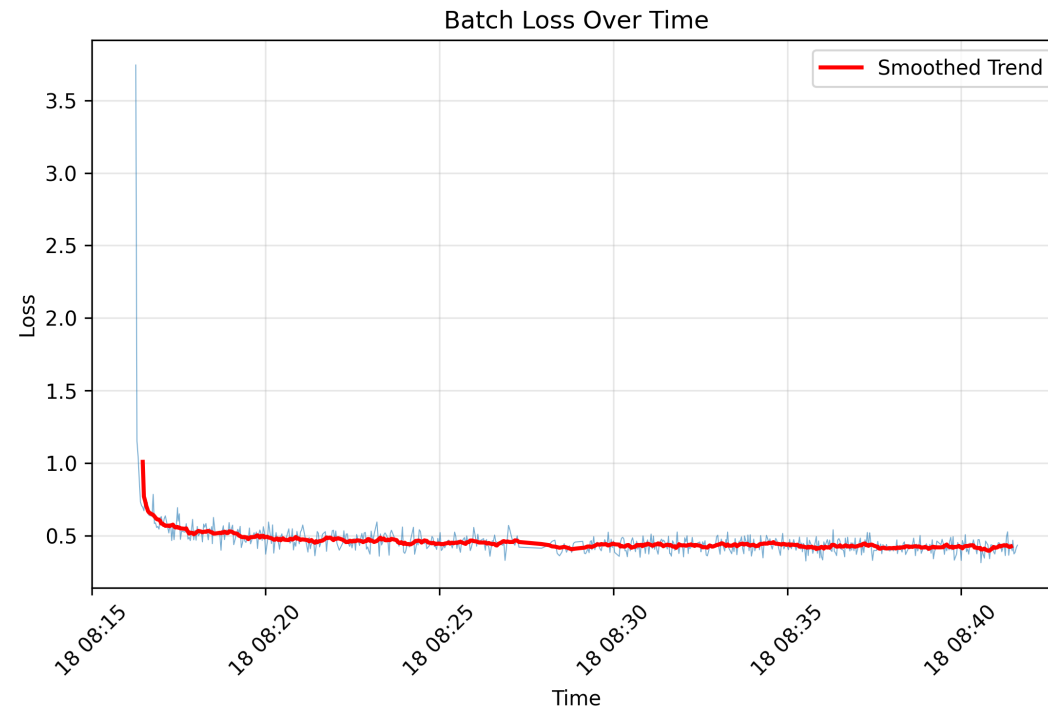
M

# Training Progress - Gigantic Background Experiment



S

# Training Progress - Gigantic Background Experiment

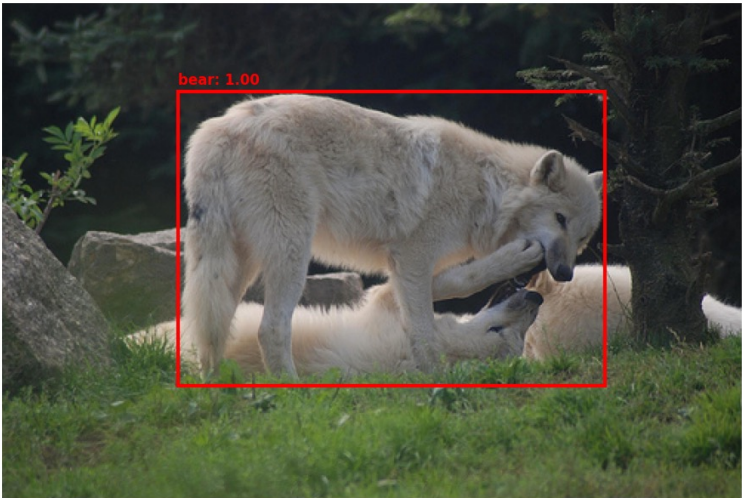


# Zero Shot Segmentation

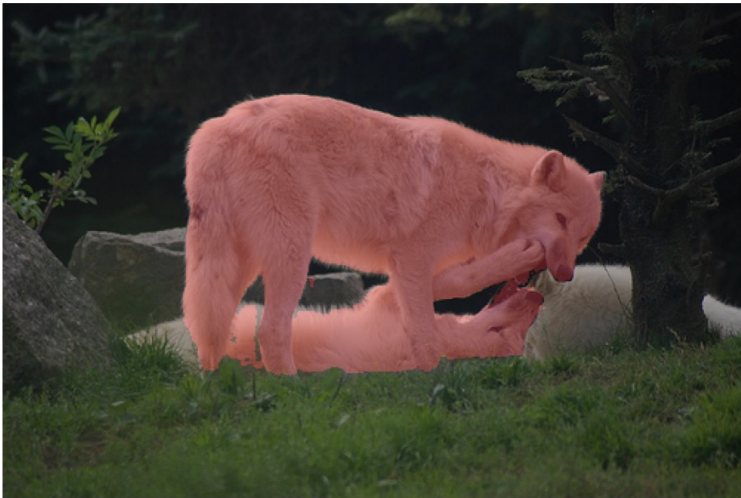
Original Image



Object Detection (Bounding Boxes)



Segmentation Masks (Basic)



Original Image



First Mask (Raw Values)  
Min: 0.000, Max: 1.000



First Mask (Thresholded)  
Non-zero pixels: 40340



# MAE Impl

Original Image



Segmentation Masks



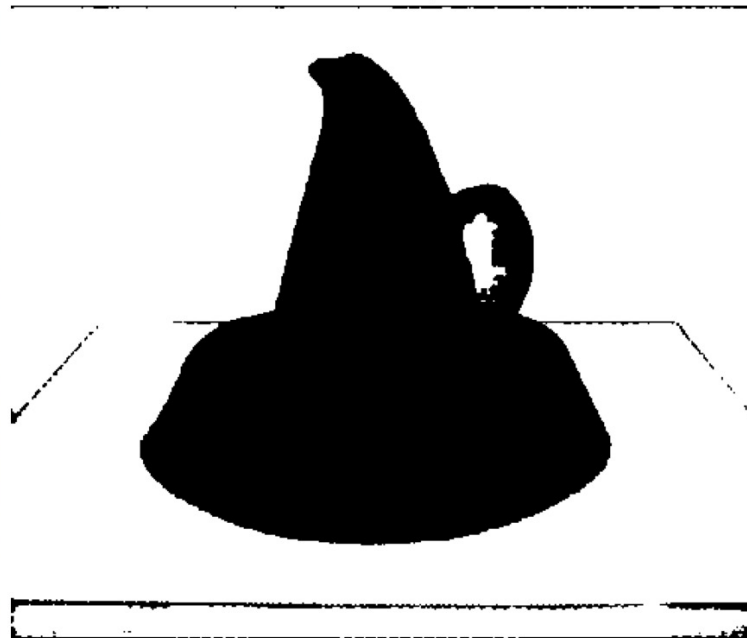
Labeled Segments



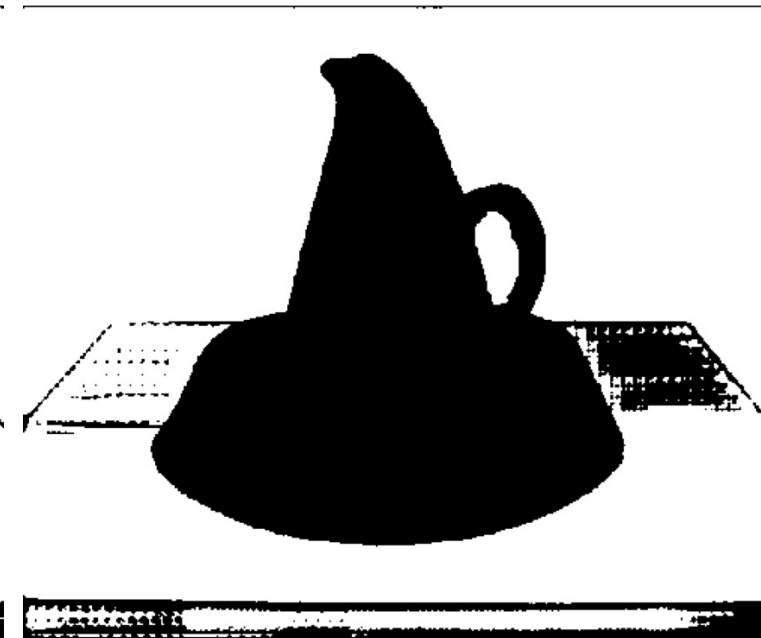
Original Image



First Segment Mask  
Area: 149561 pixels



Second Segment Mask  
Area: 140304 pixels



Original Image



Segmentation Masks



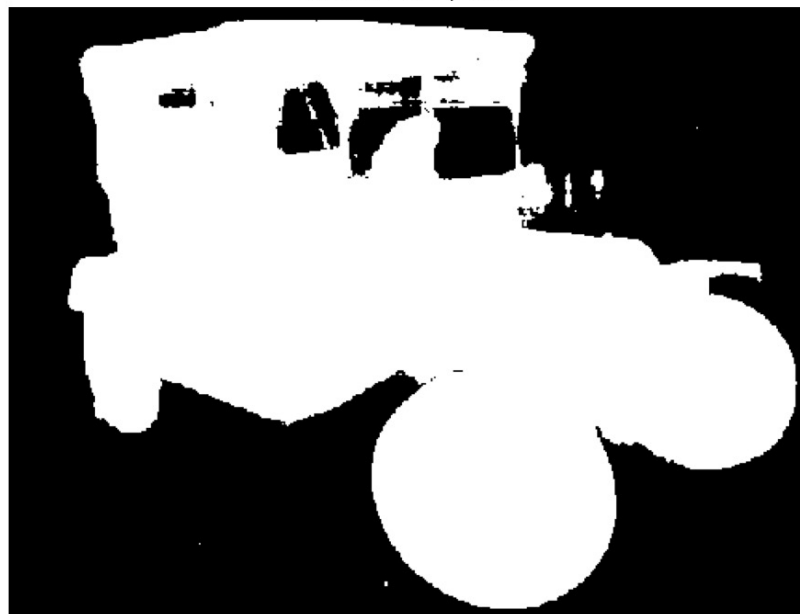
Labeled Segments



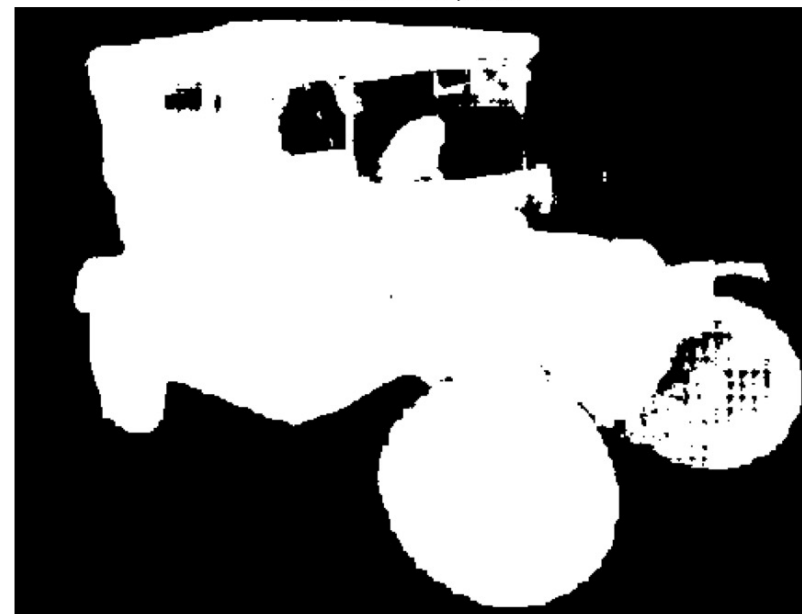
Original Image



First Segment Mask  
Area: 94494 pixels



Second Segment Mask  
Area: 90428 pixels



# Testing ViT(Vision Transformers)

- Attempted to try a zero-shot segmentation algorithm
  - Used OpenAI CLIP
  - Grounding DINO
  - Meta's SAM
- Too much effort for something that might not pan out – could be future work